

DRAWING LIST FOR TEST PACKAGE

IW-528

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	528	5FA	20A-IW-ST1-C-9741-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
2	528	5FA	20A-IW-ST1-C-9742-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
3	528	5FA	20A-IW-ST1-C-9743-01	1 OF 1	0	CJ-M15W14-PR-A20-000103	1A	
4	528	5FA	20A-IW-ST1-C-9744-01	1 OF 1	0	CJ-M15W14-PR-A20-000103	1A	
5	528	5FA	20A-IW-ST1-C-9745-01	1 OF 1	0	CJ-M15W14-PR-A20-000104	1A	
6	528	5FA	20A-IW-ST1-C-9746-01	1 OF 1	0	CJ-M15W14-PR-A20-000104	1A	
7	528	5FA	20A-IW-ST1-C-9747-01	1 OF 1	0	CJ-M15W14-PR-A20-000105	1A	
8	528	5FA	20A-IW-ST1-C-9748-01	1 OF 1	0	CJ-M15W14-PR-A20-000105	1A	
9	528	5FA	20A-IW-ST1-C-9749-01	1 OF 1	0	CJ-M15W14-PR-A20-000107	1A	
10	528	5FA	20A-IW-ST1-C-9750-01	1 OF 1	0	CJ-M15W14-PR-A20-000107	1A	
11	528	5FA	20A-IW-ST1-C-9751-01	1 OF 1	0	CJ-M15W14-PR-A20-000108	1A	
12	528	5FA	20A-IW-ST1-C-9752-01	1 OF 1	0	CJ-M15W14-PR-A20-000108	1A	
13	528	5FA	20A-IW-ST1-C-9753-01	1 OF 1	0	CJ-M15W14-PR-A20-000109	1A	
14	528	5FA	20A-IW-ST1-C-9754-01	1 OF 1	0	CJ-M15W14-PR-A20-000109	1A	
15	528	5FA	20A-IW-ST1-C-9755-01	1 OF 1	0	CJ-M15W14-PR-A20-000110	1A	
16	528	5FA	20A-IW-ST1-C-9756-01	1 OF 1	0	CJ-M15W14-PR-A20-000110	1A	
17	528	5FA	25A-IW-ST1-C-9893-01	1 OF 1	0	CJ-M15W13-PR-A20-000081-A	2	
18	528	5FA	25A-IW-ST1-C-9894-01	1 OF 1	0	CJ-M15W13-PR-A20-000081-A	2	
19	528	5FA	25A-IW-ST1-C-9895-01	1 OF 1	0	CJ-M15W13-PR-A20-000081-A	2	
20	528	5FA	40A-IW-ST1-C-9525-01	1 OF 6	0	CJ-M15W14-PR-A20-000142	1A	

DRAWING LIST FOR TEST PACKAGE**IW-528**

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	

DRAWING LIST FOR TEST PACKAGE**IW-528**

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
21	528	5FA	40A-IW-ST1-C-9525-02	2 OF 6	0	CJ-M15W14-PR-A20-000142	1A	
22	528	5FA	40A-IW-ST1-C-9525-03	3 OF 6	0	CJ-M15W14-PR-A20-000142	1A	
23	528	5FA	40A-IW-ST1-C-9525-04	4 OF 6	0	CJ-M15W14-PR-A20-000142	1A	
24	528	5FA	40A-IW-ST1-C-9525-05	5 OF 6	0	CJ-M15W14-PR-A20-000142	1A	
25	528	5FA	40A-IW-ST1-C-9525-06	6 OF 6	0	CJ-M15W14-PR-A20-000142	1A	
26	528	5FA	40A-IW-ST1-C-9534-01	1 OF 2	0	CJ-M15W14-PR-A20-000102	1A	
27	528	5FA	40A-IW-ST1-C-9534-02	2 OF 2	0	CJ-M15W14-PR-A20-000102	1A	
28	528	5FA	40A-IW-ST1-C-9535-01	1 OF 2	0	CJ-M15W14-PR-A20-000103	1A	
29	528	5FA	40A-IW-ST1-C-9535-02	2 OF 2	0	CJ-M15W14-PR-A20-000103	1A	
30	528	5FA	40A-IW-ST1-C-9536-01	1 OF 2	0	CJ-M15W14-PR-A20-000104	1A	
31	528	5FA	40A-IW-ST1-C-9536-02	2 OF 2	0	CJ-M15W14-PR-A20-000104	1A	
32	528	5FA	40A-IW-ST1-C-9537-01	1 OF 2	0	CJ-M15W14-PR-A20-000105	1A	
33	528	5FA	40A-IW-ST1-C-9537-02	2 OF 2	0	CJ-M15W14-PR-A20-000105	1A	
34	528	5FA	40A-IW-ST1-C-9565-01	1 OF 2	0	CJ-M15W14-PR-A20-000107	1A	
35	528	5FA	40A-IW-ST1-C-9565-02	2 OF 2	0	CJ-M15W14-PR-A20-000107	1A	

DRAWING LIST FOR TEST PACKAGE**IW-528**

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

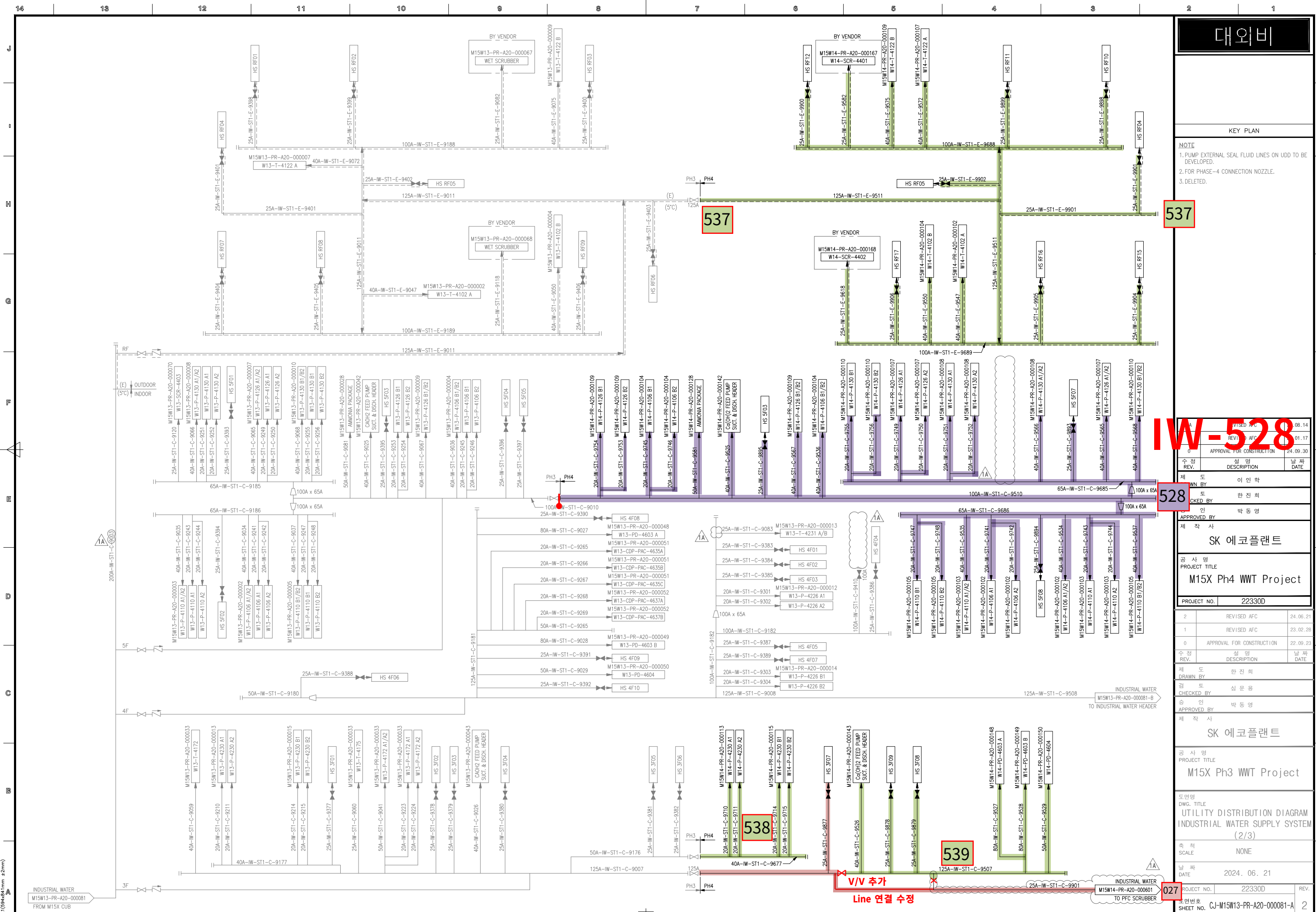
No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	

DRAWING LIST FOR TEST PACKAGE**IW-528**

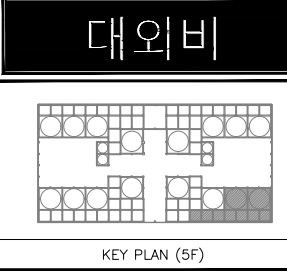
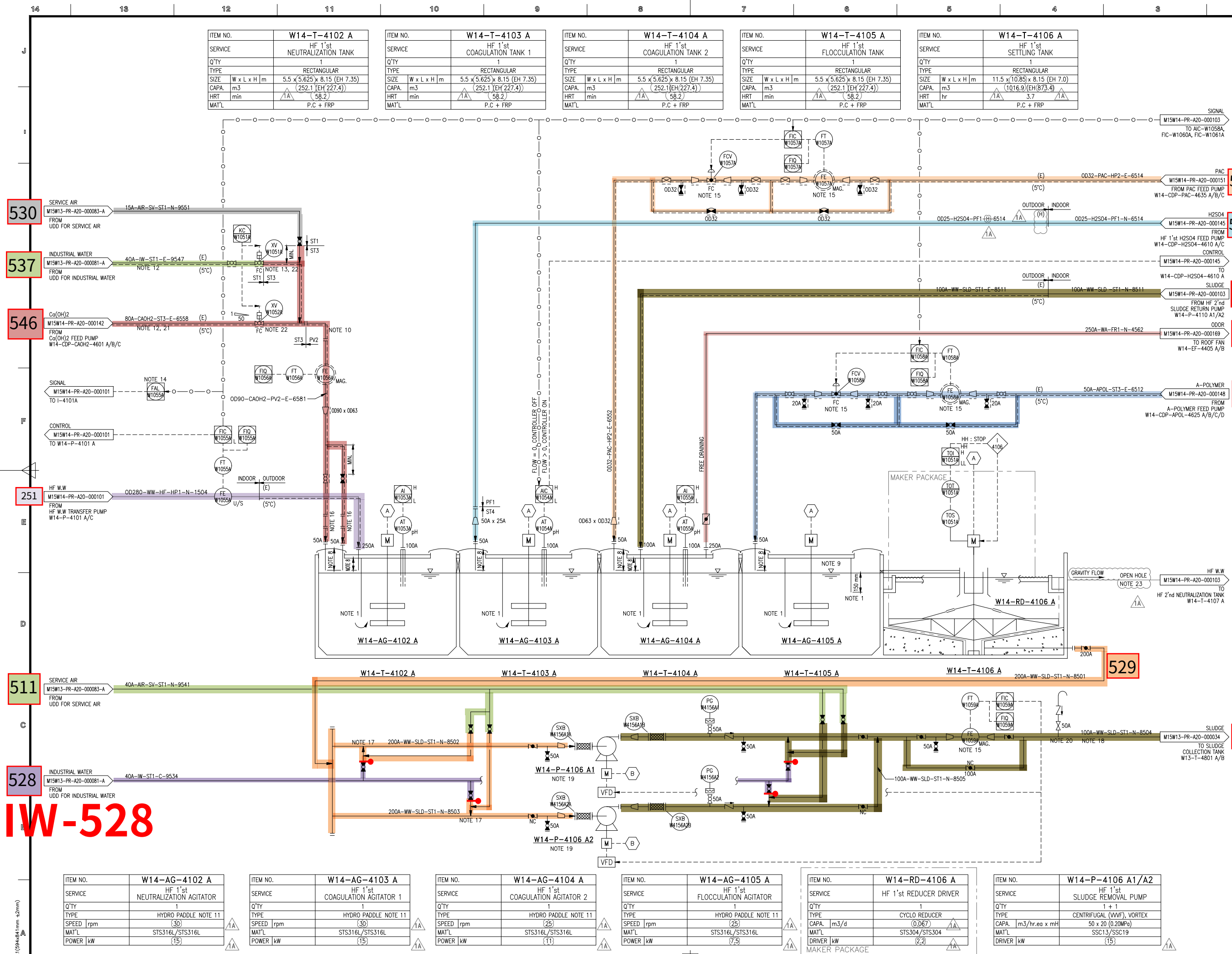
PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
36	528	5FA	40A-IW-ST1-C-9566-01	1 OF 2	0	CJ-M15W14-PR-A20-000108	1A	
37	528	5FA	40A-IW-ST1-C-9566-02	2 OF 2	0	CJ-M15W14-PR-A20-000108	1A	
38	528	5FA	40A-IW-ST1-C-9567-01	1 OF 2	0	CJ-M15W14-PR-A20-000109	1A	
39	528	5FA	40A-IW-ST1-C-9567-02	2 OF 2	0	CJ-M15W14-PR-A20-000109	1A	
40	528	5FA	40A-IW-ST1-C-9568-01	1 OF 2	0	CJ-M15W14-PR-A20-000110	1A	
41	528	5FA	40A-IW-ST1-C-9568-02	2 OF 2	0	CJ-M15W14-PR-A20-000110	1A	
42	528	5FA	50A-IW-ST1-C-9581-01	1 OF 1	0	CJ-M15W14-PR-A20-000128	1A	
43	528	5FA	65A-IW-ST1-C-9685-01	1 OF 1	0	CJ-M15W13-PR-A20-000081-A	2	
44	528	5FA	65A-IW-ST1-C-9686-01	1 OF 1	0	CJ-M15W13-PR-A20-000081-A	2	
45	528	5FA	100A-IW-ST1-C-9510-01	1 OF 1	0	CJ-M15W13-PR-A20-000081-A	2	
46	528	RFA	50A-IW-ST1-E-9581-01	1 OF 1	1	CJ-M15W14-PR-A20-000128	1A	



대외비			
KEY PLAN			
NOTE			
1. PUMP EXTERNAL SEAL FLUID LINES ON UDD TO BE DEVELOPED.			
2. FOR PHASE-4 CONNECTION NOZZLE.			
3. DELETED.			
537			
528			
IW-528			
538			
539			
V/V 추가 Line 연결 수정			
027			
INDUSTRIAL WATER M15W13-PR-A20-000081-B TO INDUSTRIAL WATER HEADER			
INDUSTRIAL WATER M15W14-PR-A20-000061 TO PFC SCRUBBER			
PROJECT NO. 22330D			
REV. 2			
SHEET NO. CJ-M15W13-PR-A20-000081-A			



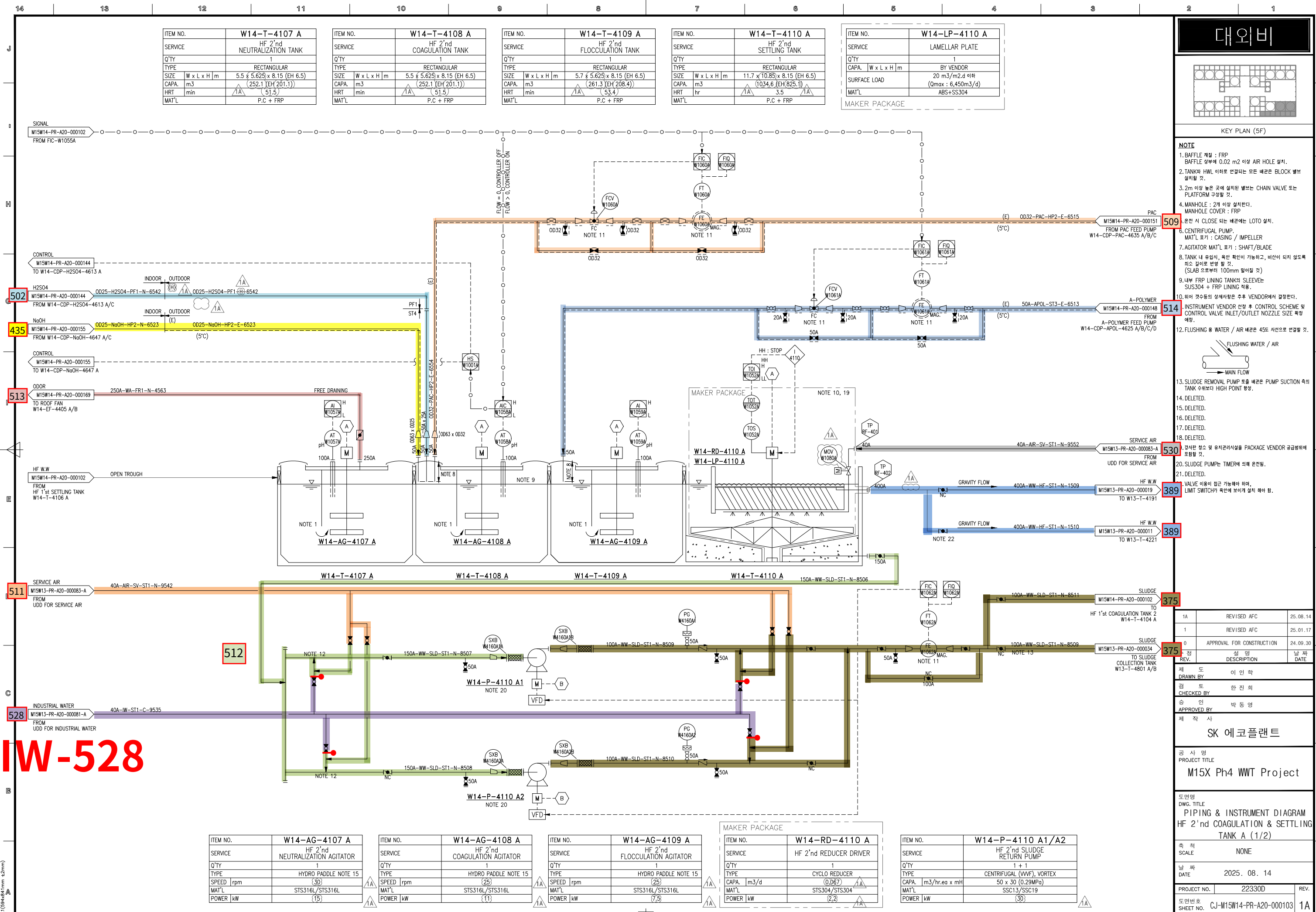
- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치할 것.
 2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.
 3. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
후진 시 CLOSE 되는 배관에는 LOTO 설치.
 4. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 5. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 6. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 반영 할 것.
 7. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
SS304 + FRP LINING 적용.
 8. 11. VENDOR에서 결정한다.
 9. 12. 라인은 상부에서 시수 도입하도록 설치한다.
W1051A OPEN (12'간마다 15') 일 경우 :
XV-W1052A CLOSE
 10. 14. W14-P-4101 A가 RUN 후 30초 후부터 동작한다.
 11. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 12. 16. 구간 투명 PVC 적용. (주입 여부 확인용)
 13. 17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 14. 18. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
 15. 19. SLUDGE PUMP는 TIMER에 의해 운전됨.
 16. 20. HIGH POINT에 SIPHON BREAKER 설치할것.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
 17. 21. 소액의 공금 HEADER에서 주입되는 POINT 별
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 위험 최소화
및 주입배관 고제를 용이하게 한다.
 18. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 19. 23. CONCRETE WALL OPENING을 통한 유체 이용
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제	도	이	인	학
DRAWN BY				
검	토	한	진	회
CHECKED BY				
인	박	동	영	
APPROVED BY				
제	작	사		

SK 에코플랜트	
공 사 명	
PROJECT TITLE	
M15X Ph4 WWT Project	
도면명	
DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM	
HF 1st COAGULATION & SETTLING	
TANK A (1/2)	
속	척
SCALE	NONE
날	짜
DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000102
SHEET NO.	1A

IW-528



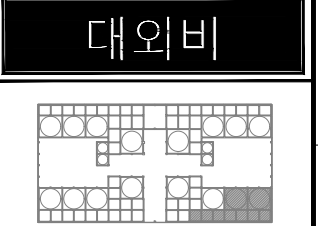
ITEM NO.	W14-T-4107 A	
SERVICE	HF 2'nd NEUTRALIZATION TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	5.5 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3	(252.1 (EH 201.1))
HRT	min	(51.5)
MAT'L	P.C + FRP	

ITEM NO.	W14-T-4108 A	
SERVICE	HF 2'nd COAGULATION TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	5.5 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3	(252.1 (EH 201.1))
HRT	min	(51.5)
MAT'L	P.C + FRP	

ITEM NO.	W14-T-4109 A	
SERVICE	HF 2'nd FLOCCULATION TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	5.7 x 5.625 x 8.15 (EH 6.5)
CAPA.	m3	(261.3 (EH 208.4))
HRT	min	(53.4)
MAT'L	P.C + FRP	

ITEM NO.	W14-T-4110 A	
SERVICE	HF 2'nd SETTLING TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	11.7 x 10.85 x 8.15 (EH 6.5)
CAPA.	m3	(1034.6 (EH 825.1))
HRT	hr	3.5
MAT'L	P.C + FRP	

ITEM NO.	W14-LP-4110 A	
SERVICE	LAMELLAR PLATE	
Q'TY	1	
CAPA.	W x L x H m	BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)	
MAT'L	ABS+SS304	



- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB 으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 것수등의 상세사항은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 검사한 장소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.
 22. VALVE 이용이 접근 가능해야 하며, LIMIT SWITCH가 육안에 보이기 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
REV.	실명	날짜
DESCRIPTION		DATE

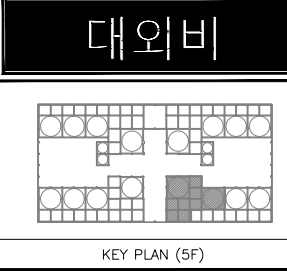
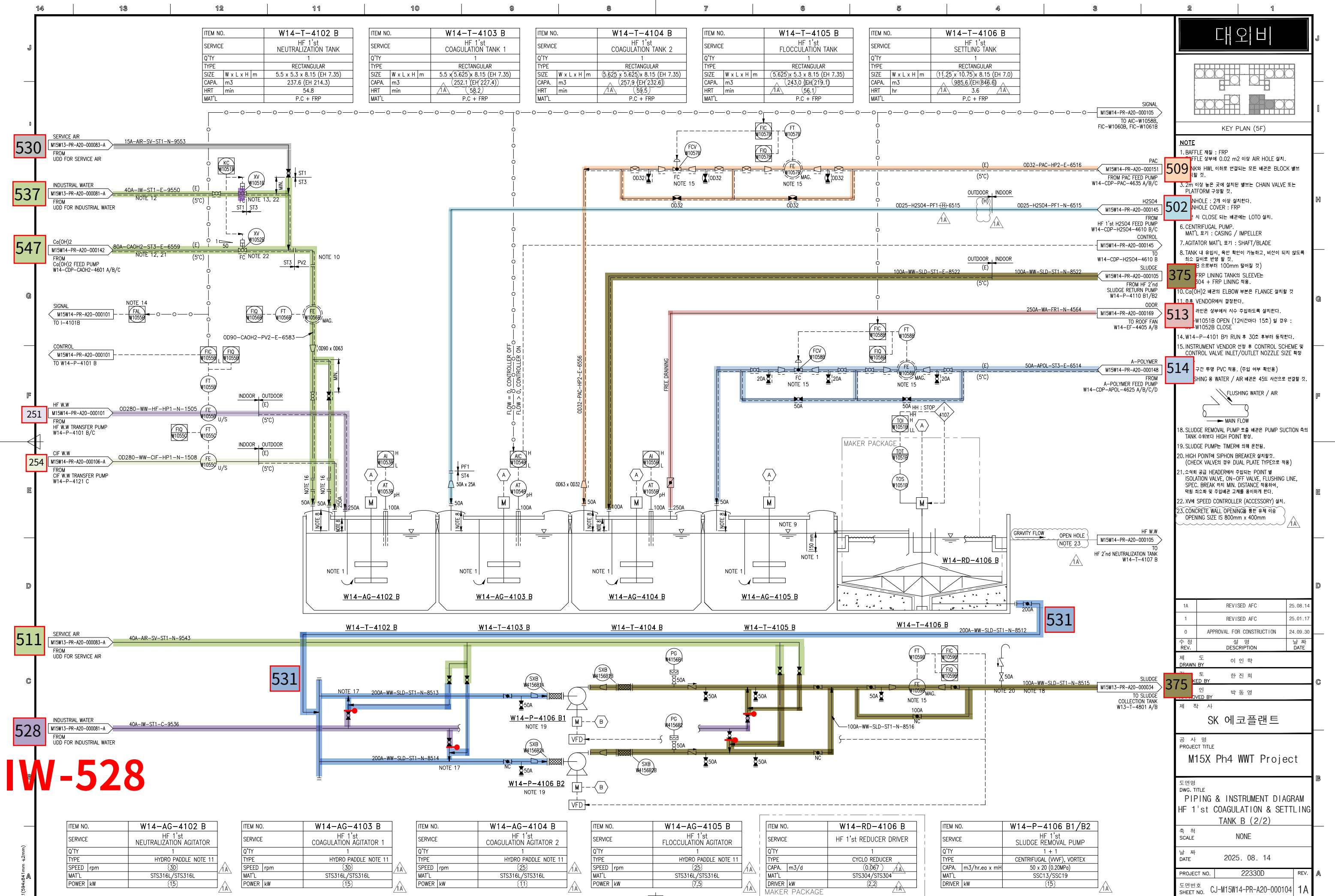
제 도	이 인 학
검 토	한 진 희
CHECKED BY	
승 인	박 동 명
APPROVED BY	
제 작 사	

SK 에코플랜트	
공 사 명	PROJECT TITLE
M15X Ph4 WWT Project	

도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM	
HF 2'nd COAGULATION & SETTLING TANK A (1/2)	
속 척	SCALE
NONE	
날 짜	DATE
2025. 08. 14	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000103
SHEET NO.	1A

IW-528

IW-528



- NOTE
1. Baffle 제형 : FRP
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치.
 3. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.
 4. INHOLE : 2개 이상 설치한다.
INHOLE COVER : FRP
시 CLOSE 되는 배관에는 LOTO 설치.
 5. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 할 것.
B 으로부터 100mm 떨어질 것
 9. FRP LINING TANK의 SLEEVE는
504 + FRP LINING 적용.
 10. Ca(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
 11. 주용 VENDOR에서 결정한다.
 12. 라인은 상부에서 시공 시 유입하드록 설치한다.
 13. W1051B OPEN (12시간마다 15초) 일 경우 :
W1052B CLOSE
 14. W14-P-4101 B가 RUN 후 30초 후부터 동작한다.
 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 16. 구간 투명 PVC 적용. (주입 여부 확인용)
 17. SHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
-
- FLUSHING WATER / AIR
MAIN FLOW

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	인
DRAWN BY	이인학	
토	한진희	
ED BY	박동영	
제	작	사

SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
HF 1'st COAGULATION & SETTLING		
TANK B (2/2)		
속	척	
SCALE	NONE	
날	짜	
DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000104	1A

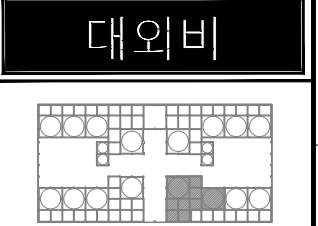
ITEM NO.	W14-T-4107 B
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.45 x 8.15 (EH 6.5) (255.4 (EH 203.7)
CAPA.	m3
HRT	min
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 B
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.175 x 8.15 (EH 6.5) (242.5 (EH 193.4)
CAPA.	m3
HRT	min
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 B
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.625 x 5.175 x 8.15 (EH 6.5) (237.2 (EH 189.2)
CAPA.	m3
HRT	min
MAT'L	P.C + FRP

ITEM NO.	W14-T-4110 B
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (11.375 x 10.85 x 8.15 (EH 6.5) (1005.9 (EH 802.2)
CAPA.	m3
HRT	hr
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 B
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304



KEY PLAN (5F)

- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
HOLE COVER : FRP

5. 시 CLOSE 되는 배관에는 LOTO 설치.
6. TRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB 으로부터 100mm 떨어질 것)
9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
10. 특수용의 설계사항은 추후 VENDOR에서 결정한다.
11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.

13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
14. DELETED.
15. DELETED.
16. DELETED.
17. DELETED.
18. DELETED.
19. DELETED.
20. SLUDGE PUMP는 TIMER에 의해 운전됨.
21. DELETED.

22. 용이 접근 가능해야 하며, ITCH가 육안에 보이게 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
375	APPROVAL FOR CONSTRUCTION	24.09.30
375	실명 DESCRIPTION	날짜 DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
승 인	박 동 명
APPROVED BY	
제 작 사	

공 사 명	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM	
HF 2'nd COAGULATION & SETTLING TANK B (2/2)	
속 척	SCALE
NONE	
날 짜	DATE
2025. 08. 14	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000105
SHEET NO.	1A

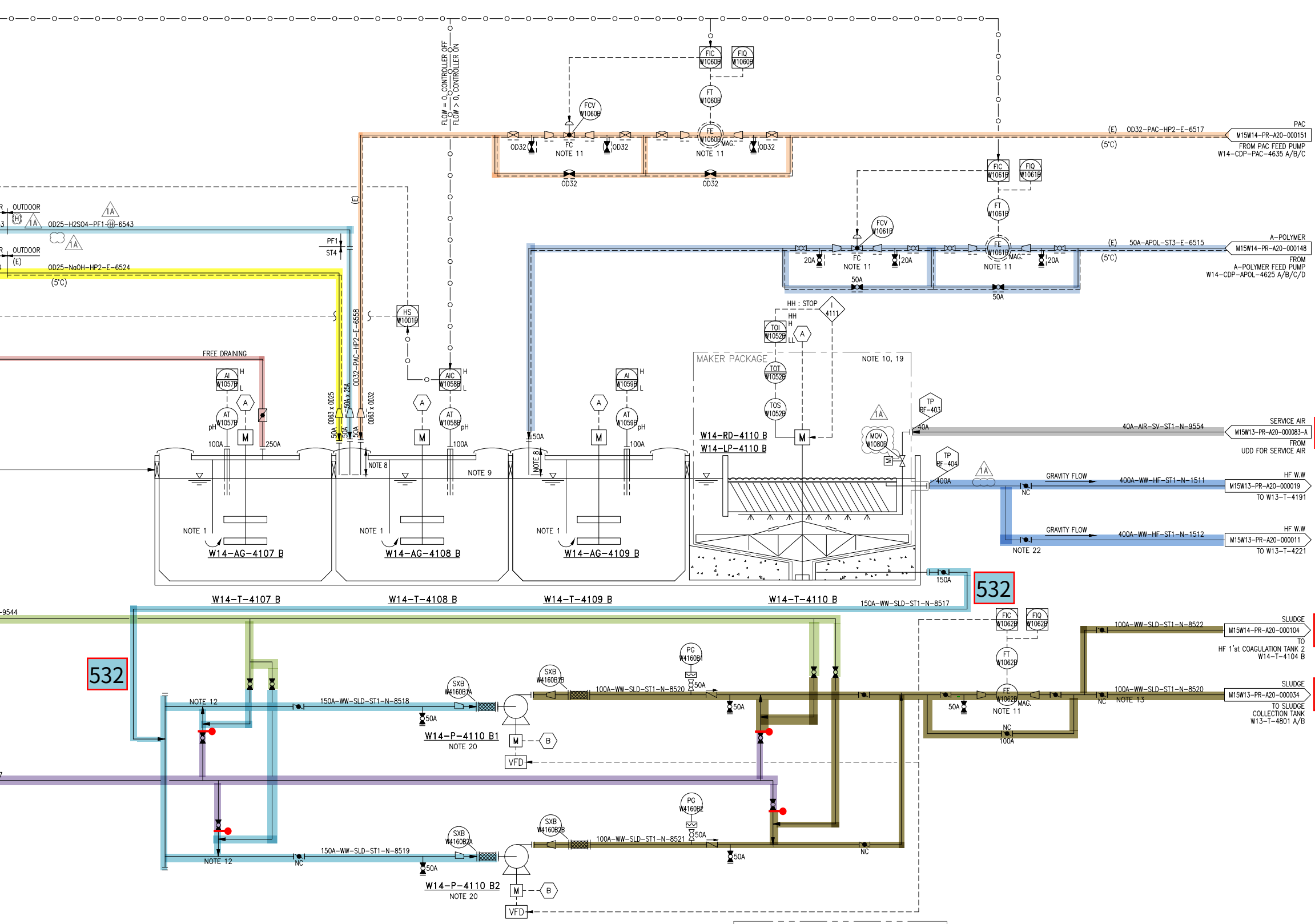
ITEM NO.	W14-AG-4107 B
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm
(30)	
MAT'L	STS316L/STS316L
POWER	kw
(15)	

ITEM NO.	W14-AG-4108 B
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm
(25)	
MAT'L	STS316L/STS316L
POWER	kw
(15)	

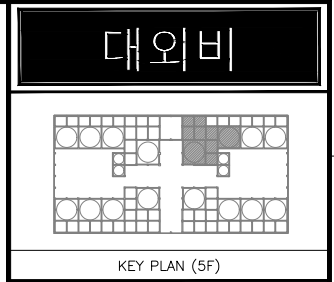
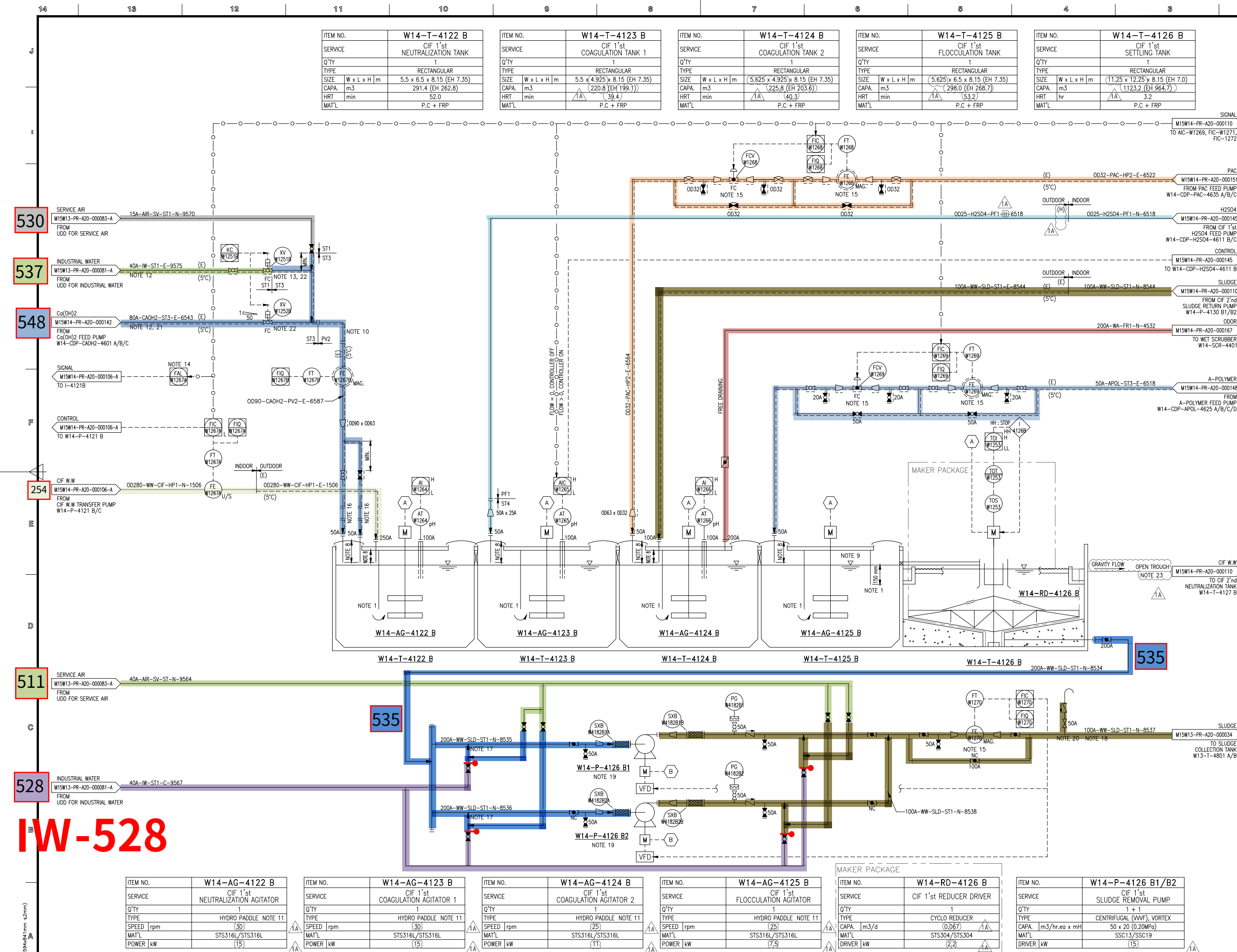
ITEM NO.	W14-AG-4109 B
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm
(25)	
MAT'L	STS316L/STS316L
POWER	kw
(7.5)	

ITEM NO.	W14-RD-4110 B
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d
(0.067)	
MAT'L	STS304/STS304
POWER	kw
(2.2)	

ITEM NO.	W14-P-4110 B1/B2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH
50 x 30 (0.29MPa)	
MAT'L	SSC13/SSC19
POWER	kw
(30)	



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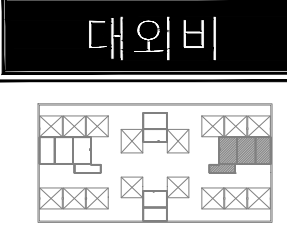
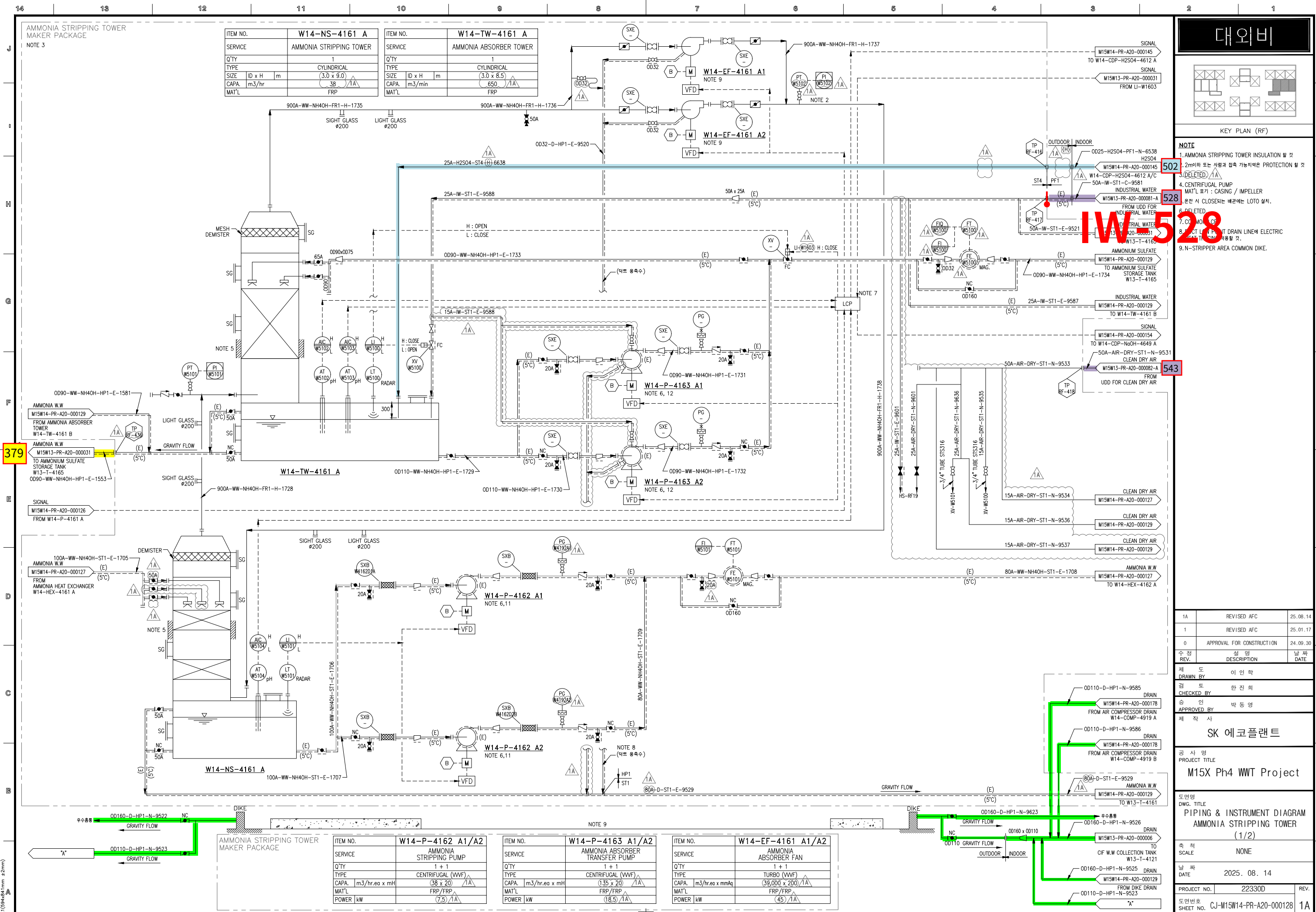


- NOTE
1. Baffle 제형 : FRP
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.
NK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
시설 함.
 2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성함.
 3. HOLE : 2개 이상 설치한다.
시 CLOSE 되는 배관에는 LOTO 설치.
 4. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 5. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 6. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 함.
 7. B로부터 100mm 떨어질 것)
 8. FRP LINING TANK의 SLEEVE는
304 + FRP LINING 적용.
 9. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치함.
 10. VENDOR에서 결정한다.
 11. 라인은 상부에서 시수 주입하도록 설치한다.
 12. W125B OPEN (12시간마다 15초) 일 경우 :
XV-W125B CLOSE
 13. 14. W14-P-4121 B가 RUN 후 30초 후부터 동작한다.
 14. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET / OUTLET NOZZLE SIZE
예정.
 15. 17. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결함.
 16. 18. SLUDGE REMOVAL PUMP 또는 배관을 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.
 17. 19. SLUDGE PUMP는 TIMER에 의해 운전됨.
 18. 20. HIGH POINT에 SIPHON BREAKER 설치함.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
 19. 21. 소액의 공구 HEADER에서 주입되는 POINT
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여,
역침 침소화 및 주입배관 고체를 용이하게 한다.
 20. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 21. 23. CONCRETE WALL OPENING을 통한 유체 이용
OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

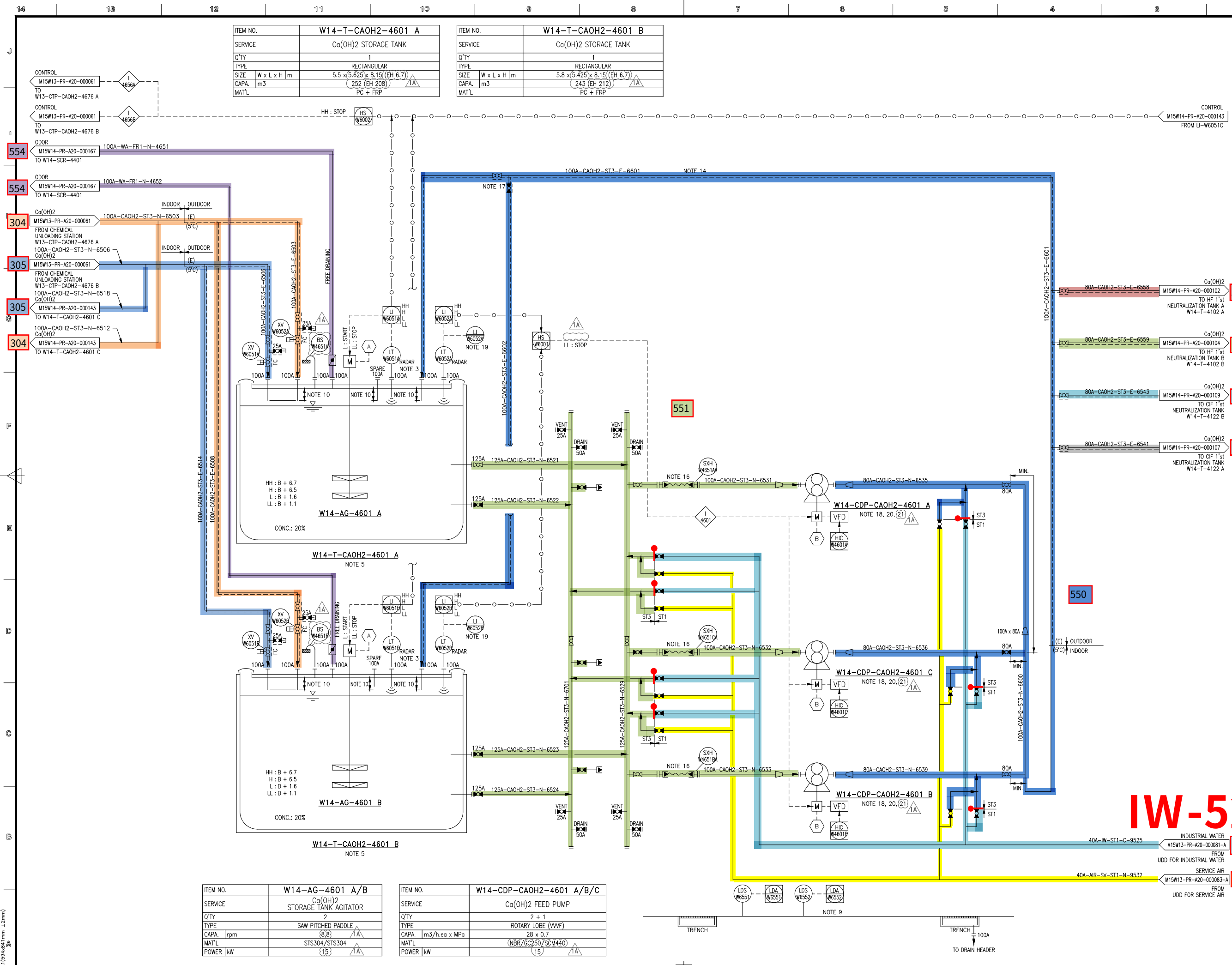
제도	이인학
DRAWN BY	
검토	한진희
CHECKED BY	
인	박동영
APPROVED BY	
제작사	

SK 에코플랜트	
공사명	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	DWG. TITLE
PIPING & INSTRUMENT BIAGRAM CIF 1'st COAGULATION & SETTLING TANK B (2/2)	
속척	SCALE
NONE	
날짜	DATE
2025. 08. 14	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000109
SHEET NO.	1A



- NOTE
- 1. AMMONIA STRIPPING TOWER INSULATION 및 2. 2m이하 또는 사면과 접촉 가능지역은 PROTECTION 할 것
 - 3. ~~DELETED~~ / 1A
 - 4. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
 - 5. 본선 시 CLOSE되는 배관에는 LOTO 설치
 - 6. ~~DELETED~~
 - 7. CO. NO. OF
 - 8. FIRST LIFT POINT DRAIN LINE에 ELECTRIC
 - 9. N-STRIPPER AREA COMMON DIKE.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE
제도 DRAWN BY	이인학	
검토 CHECKED BY	한진희	
승인 APPROVED BY	박동명	
제작사	SK 에코플랜트	
공사명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM AMMONIA STRIPPING TOWER (1/2)	
속척 SCALE	NONE	
날짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV. A
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000128	1A



ITEM NO.	W14-T-CAOH2-4601 A
SERVICE	Ca(OH)2 STORAGE TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.7)
CAPA.	m3 (252 (EH 208))
MAT'L	PC + FRP

ITEM NO.	W14-T-CAOH2-4601 B
SERVICE	Ca(OH)2 STORAGE TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.8 x 5.425 x 8.15 (EH 6.7)
CAPA.	m3 (243 (EH 212))
MAT'L	PC + FRP

ITEM NO.	W14-AG-4601 A/B
SERVICE	Ca(OH)2 STORAGE TANK AGITATOR
Q'TY	2
TYPE	SAW PITCHED PADDLE
CAPA.	rpm (8.8) 1A
MAT'L	STS304/STS304
POWER	kW (15) 1A

ITEM NO.	W14-CDP-CAOH2-4601 A/B/C
SERVICE	Ca(OH)2 FEED PUMP
Q'TY	2 + 1
TYPE	ROTARY LOBE (VWF)
CAPA.	m3/h.ea x MPa 28 x 0.7
MAT'L	(NBR/GC250/SCM440)
POWER	kW (15) 1A

대외비

KEY PLAN (5F)

NOTE

1. CaOH2 배관의 ELBOW 부분은 FLANGE 설치할 것.

2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.

3. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.

4. P&ID에 표기된 모든 ELECTRIC POWER는 VENDOR PRINT 참조 후 작성할 것.

5. 5층에 설치할 것.

6. 약품종류에 따라 LINE 색상을 구분하여야 한다.

7. 운전 시 CLOSE 되는 배관에는 LOTO 설치.

8. AGITATOR MAT'L 표기 : SHAFT/BLADE

9. CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 발생되는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.

10. TANK 내 유압, 폭탄 확인이 가능하고, 비산되지 않도록 최소 길이로 반영할 것 (SLAB로부터 100mm 떨어진 것).

11. MANHOLE : 2개 이상 설치한다.

12. INTERLOCK NO. 1-4601 (W14-CDP-CAOH2-4601 A/B/C AUTO STOP) : REMOTE STATUS# W14-T-CAOH2-4601 A/B LEVEL (LI-W6052A/B) LOW LOW STOP.

13. INTERLOCK NO. 1-4656A/B (W13-CTP-CAOH2-4676 A/B AUTO STOP) : REMOTE STATUS# W14-T-CAOH2-4601 A/B/C LEVEL (LI-W6051A/B/C) HIGH HIGH STOP.

14. RINGMAN LINE SCHEMATIC

SCHEMATIC DRAWING

15. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.

16. 7개 VENDOR LOOSE PART 공급.

17. 운전중 Ca(OH)2 TANK A/B RETURN VALVE 2ea 중 1ea는 반드시 OPEN 되어있어야 한다.

18. INTERNAL PSV 설치 할 것.

19. 5층 소액취 주입포트 인근에 TANK LEVEL 확인 가능하도록 LOCAL PANEL 설치.

20. 소액취 주입 포트 AREA에는 비산 방지막을 설치한다.

21. INTERNAL PSV SET PRESSURE : 0.8 Mpa(g)

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	인
DRAWN BY	이인학	
검	토	한진희
CHECKED BY		
승	인	박동영
APPROVED BY		
제	작	사
SK 에코플랜트		
PROJECT TITLE		
13X Ph4 WWT Project		
28	면명	
WG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
Ca(OH)2 STORAGE TANK & FEED PUMP (1/2)		
SCALE		
NONE		
날	짜	
DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호		
SHEET NO.	CJ-M15W14-PR-A20-000142	1A

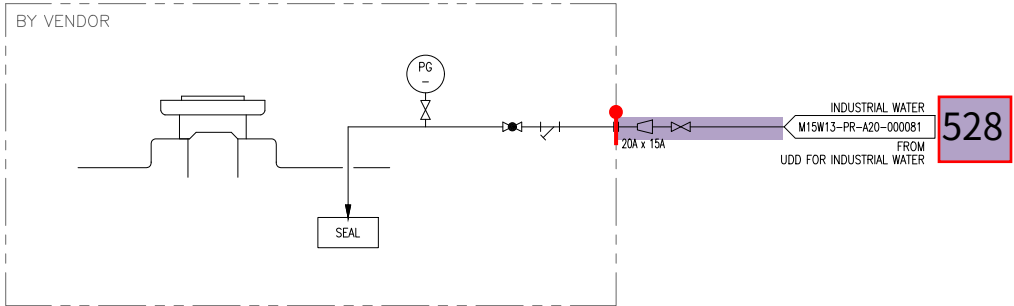
IW-528

대외비

NOTE

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 이 인 학		
DRAWN BY		
검 토 한 진 희		
CHECKED BY		
승 인 박 동 영		
APPROVED BY		
제 작 사		
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
PUMP SEAL & PIPING PLAN		
(1/2)		
축척		
SCALE		
NONE		
날짜		
DATE		
2025. 08. 14		
PROJECT NO.		22330D
도면번호		CJ-M15W14-PR-A00-000011
SHEET NO.		2A

SEAL PLAN 32
FOR EACH SEAL FLUSHING SYSTEM

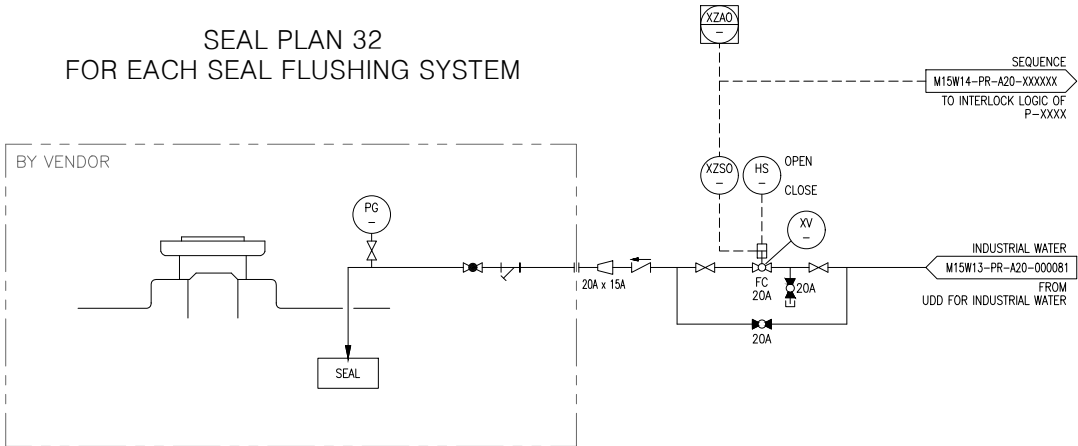


ITEM NO.	P&ID NO.	PG	EXTERNAL SEAL FLUID LINE NO.
W14-P-4106 A1	M15W14-PR-A20-000102	PG-W8156A1	20A-IW-ST1-C-9741
W14-P-4106 A2	M15W14-PR-A20-000102	PG-W8156A2	20A-IW-ST1-C-9742
W14-P-4110 A1	M15W14-PR-A20-000103	PG-W8160A1	20A-IW-ST1-C-9743
W14-P-4110 A2	M15W14-PR-A20-000103	PG-W8160A2	20A-IW-ST1-C-9744
W14-P-4106 B1	M15W14-PR-A20-000104	PG-W8156B1	20A-IW-ST1-C-9745
W14-P-4106 B2	M15W14-PR-A20-000104	PG-W8156B2	20A-IW-ST1-C-9746
W14-P-4110 B1	M15W14-PR-A20-000105	PG-W8160B1	20A-IW-ST1-C-9747
W14-P-4110 B2	M15W14-PR-A20-000105	PG-W8160B2	20A-IW-ST1-C-9748
W14-P-4121 A	M15W14-PR-A20-000106-A	PG-W8171A	20A-IW-ST1-C-9701
W14-P-4121 B	M15W14-PR-A20-000106-A	PG-W8171B	20A-IW-ST1-C-9702
W14-P-4121 C	M15W14-PR-A20-000106-A	PG-W8171C	20A-IW-ST1-C-9703
W14-P-4126 A1	M15W14-PR-A20-000107	PG-W8176A1	20A-IW-ST1-C-9749
W14-P-4126 A2	M15W14-PR-A20-000107	PG-W8176A2	20A-IW-ST1-C-9750
W14-P-4130 A1	M15W14-PR-A20-000108	PG-W8180A1	20A-IW-ST1-C-9751
W14-P-4130 A2	M15W14-PR-A20-000108	PG-W8180A2	20A-IW-ST1-C-9752
W14-P-4126 B1	M15W14-PR-A20-000109	PG-W8176B1	20A-IW-ST1-C-9753
W14-P-4126 B2	M15W14-PR-A20-000109	PG-W8176B2	20A-IW-ST1-C-9754
W14-P-4130 B1	M15W14-PR-A20-000110	PG-W8180B1	20A-IW-ST1-C-9755
W14-P-4130 B2	M15W14-PR-A20-000110	PG-W8180B2	20A-IW-ST1-C-9756
W14-P-4221 A	M15W14-PR-A20-000111	PG-W8271A	20A-IW-ST1-C-9705
W14-P-4221 B	M15W14-PR-A20-000111	PG-W8271B	20A-IW-ST1-C-9706
W14-P-4221 C	M15W14-PR-A20-000111	PG-W8271C	20A-IW-ST1-C-9707
W14-P-4226 A1	M15W14-PR-A20-000112	PG-W8276A1	20A-IW-ST1-C-9801
W14-P-4226 A2	M15W14-PR-A20-000112	PG-W8276A2	20A-IW-ST1-C-9802
W14-P-4230 A1	M15W14-PR-A20-000113	PG-W8280A1	20A-IW-ST1-C-9710
W14-P-4230 A2	M15W14-PR-A20-000113	PG-W8280A2	20A-IW-ST1-C-9711
W14-P-4226 B1	M15W14-PR-A20-000114	PG-W8276B1	20A-IW-ST1-C-9803
W14-P-4226 B2	M15W14-PR-A20-000114	PG-W8276B2	20A-IW-ST1-C-9804

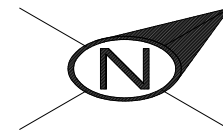
ITEM NO.	P&ID NO.	PG	EXTERNAL SEAL FLUID LINE NO.
W14-P-4230 B1	M15W14-PR-A20-000115	PG-W8280B1	20A-IW-ST1-C-9714
W14-P-4230 B2	M15W14-PR-A20-000115	PG-W8280B2	20A-IW-ST1-C-9715
W14-P-4191 A	M15W14-PR-A20-000119	PG-W8241A	20A-IW-ST1-C-9716
W14-P-4191 B	M15W14-PR-A20-000119	PG-W8241B	20A-IW-ST1-C-9717
W14-P-4191 C	M15W14-PR-A20-000119	PG-W8241C	20A-IW-ST1-C-9718
W14-P-4195 A1	M15W14-PR-A20-000120	PG-W8245A1	20A-IW-ST1-C-9719
W14-P-4195 A2	M15W14-PR-A20-000120	PG-W8245A2	20A-IW-ST1-C-9720
W14-P-4195 B1	M15W14-PR-A20-000121	PG-W8245B1	20A-IW-ST1-C-9721
W14-P-4195 B2	M15W14-PR-A20-000121	PG-W8245B2	20A-IW-ST1-C-9722
W14-P-4801 A	M15W14-PR-A20-000134	PG-W8551A	20A-IW-ST1-C-9727
W14-P-4801 B	M15W14-PR-A20-000134	PG-W8551B	20A-IW-ST1-C-9728
W14-P-4907 A	M15W14-PR-A20-000163	PG-W8957A	20A-IW-ST1-C-9729
W14-P-4907 B	M15W14-PR-A20-000163	PG-W8957B	20A-IW-ST1-C-9730
W14-P-4715 A	M15W14-PR-A20-000501	PG-W8715A	20A-IW-ST1-C-7008
W14-P-4715 B	M15W14-PR-A20-000501	PG-W8715B	20A-IW-ST1-C-7009
W14-P-4715 C	M15W14-PR-A20-000501	PG-W8715C	20A-IW-ST1-C-7010
W14-P-4725 A	M15W14-PR-A20-000502	PG-W8725A	20A-IW-ST1-C-7011
W14-P-4725 B	M15W14-PR-A20-000502	PG-W8725B	20A-IW-ST1-C-7012
W14-P-4729 A	M15W14-PR-A20-000503	PG-W8729A	20A-IW-ST1-C-7013
W14-P-4729 B	M15W14-PR-A20-000503	PG-W8729B	20A-IW-ST1-C-7014
W14-P-4730 A	M15W14-PR-A20-000504	PG-W8730A	20A-IW-ST1-C-7015
W14-P-4730 B	M15W14-PR-A20-000504	PG-W8730B	20A-IW-ST1-C-7016
W14-P-4735 A	M15W14-PR-A20-000505	PG-W8735A	20A-IW-ST1-C-7017
W14-P-4735 B	M15W14-PR-A20-000505	PG-W8735B	20A-IW-ST1-C-7018
W14-P-4738 A	M15W14-PR-A20-000506	PG-W8738A	20A-IW-ST1-C-7019
W14-P-4738 B	M15W14-PR-A20-000506	PG-W8738B	20A-IW-ST1-C-7020
W14-P-4731 A1/A2	M15W14-PR-A20-000113		
W14-P-4731 B1/B2	M15W14-PR-A20-000115		
W14-P-4232 A/B	M15W14-PR-A20-000116		
W14-P-4739 A/B	M15W14-PR-A20-000506		
W14-P-4740 A/B	M15W14-PR-A20-000507		

2A

SEAL PLAN 32
FOR EACH SEAL FLUSHING SYSTEM



ITEM NO.	P&ID NO.	PG	XV	HS	XZSO	XZAO	EXTERNAL SEAL FLUID LINE NO.
W14-P-4805 A1	M15W14-PR-A20-000135	PG-W8855A1	XV-W4855A1	HS-W4855A1	XZSO-W4855A1	XZAO-W4855A1	20A-IW-ST1-C-9737
W14-P-4805 A2	M15W14-PR-A20-000135	PG-W8855A2	XV-W4855A2	HS-W4855A2	XZSO-W4855A2	XZAO-W4855A2	20A-IW-ST1-C-9738
W14-P-4805 B1	M15W14-PR-A20-000136	PG-W8855B1	XV-W4855B1	HS-W4855B1	XZSO-W4855B1	XZAO-W4855B1	20A-IW-ST1-C-9739
W14-P-4805 B2	M15W14-PR-A20-000136	PG-W8855B2	XV-W4855B2	HS-W4855B2	XZSO-W4855B2	XZAO-W4855B2	20A-IW-ST1-C-9740



44

4



3

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	SH'T NO.	REV.	
20A-IW-ST1-C-9743	01 / 01	0	

TOR  SK ecoplant

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D						
						REF. P & ID CJ-M15W14-PR-A00-000011		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출될 수 없으며, 회사는 유포자의 불법적 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



6

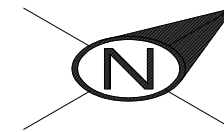
1. (*) FIELD ADJUST

CLIENT



5

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 복제, 제3자에게 공개, 누설 또는 기타 방법으로 외부에 유출된 사실이 발생하면, 회사에서는 포자의 민/형사상 책임을 물을 수 있으며, 관련 법령에서 정한 형벌을 받을 수 있습니다.



44

8



7

TOR  SK ecoplant



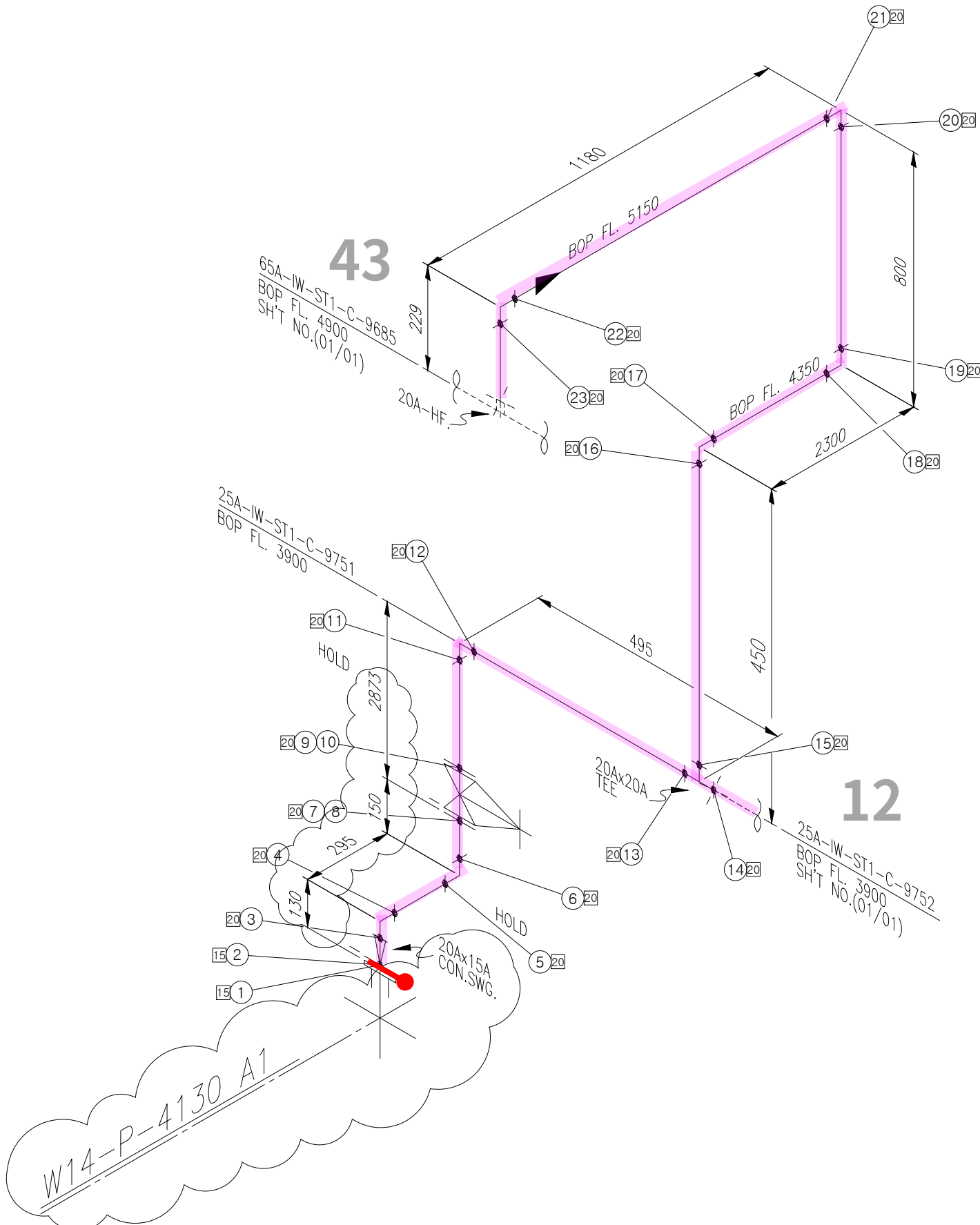
CLIENT



10

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA									M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	COMMODITY	IW	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A00-000011							REV. NO.	DATE	DESCRIPTION	BY

경고 : 퍼 도 역 SK HYNIX의 기 업 비 밀 자 료 로 서, 어 떠 한 제 3자 에 도 목 적 외 의 사 용, 복 제, 제 3자 에 게 공 개, 누 출 등 임의 의 행 위 는 엄 격히 금 지 되 며, 회 사 내 유 형 포 자 의 사 용 으 로 인 하 여 피 해 가 발 생 할 경 우 유 형 포 자 에 게 민 / 형 사 상 황 을 구 구 하 옴 수 있



<NOTE>
1. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	20A	9.0M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	20A	7EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	20A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
RCS	20Ax15A	1EA	SWAGE CON, STSF304, PBE - SCH 10S
VG	20A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STSF304, BB / OS&Y, HANDWHEEL
F	20A	2EA	FLG SO, KS 10K, RF, STS304
F	15A	1EA	FLG SO, KS 10K, RF, STS304
G	20A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	15A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	20A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	15A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
IN	25tX20A	9.0M	관 본온계(고무발포)

CLIENT



11

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
IW	22330D
LINE NO.	SH'T NO. REV.
20A-IW-ST1-C-9751	01 / 01 0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT	5FA								M15X Ph4 Project
			OPERATING	10	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-050004-D			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	IW	22330D
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO. REV.
						CJ-M15W14-PR-A00-000011								20A-IW-ST1-C-9751	01 / 01 0



CLIENT



12

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT	5FA								M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-050004-D								IW	22330D	
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A00-000011		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	20A-IW-ST1-C-9752	01 / 01	0

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 복제, 제3자에게 공개, 누설 또는 기타 방법으로 유출하여서는 안 됩니다. 이 회사는 은포지의 모든 권리를 법적으로 보호하기 위해 발생할 경우 은포지에게 민/형사상 책임을 물을 수 있습니다.

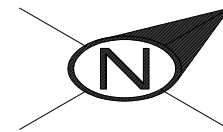


CLIENT



14

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단 복제, 배포, 제3자에게 공개, 누설될 수 있으며, 회사는 은포지의 불법 사용으로 인해 발생할 경우 은포지에게 민/형사상 책임을 요구할 수 있습니다.

[illegible]

CLIENT		 15	
CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	20A-IW-ST1-C-9755	SH'T NO. 01 / 01	REV. 0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1												
INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A00-000011						
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	



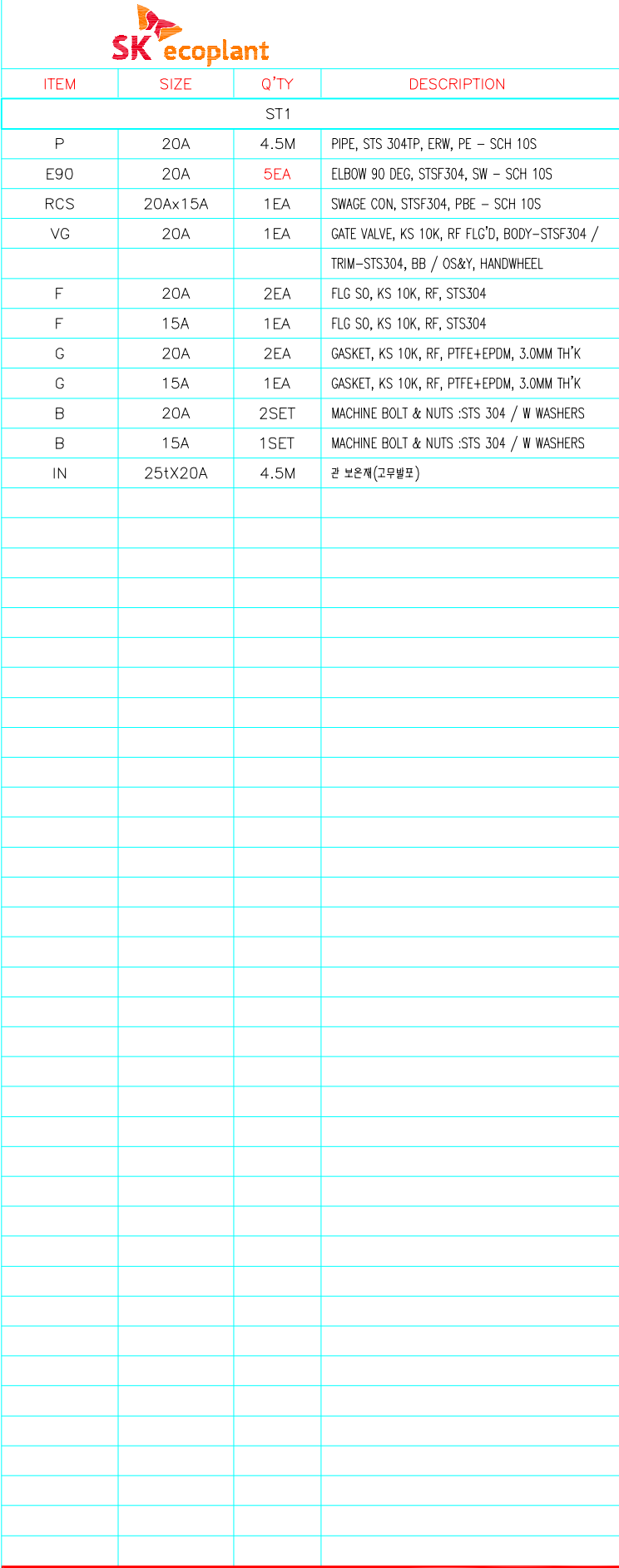
CLIENT



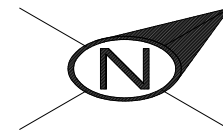
16

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	13.5	60	UNIT 5FA								M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	COMMODITY	IW	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A00-000011							REV. NO.	DATE	DESCRIPTION	BY

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있습니다.



CLIENT		16	
			
CONTRACTOR		PROJECT NAME	
			
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	20A-IW-ST1-C-9756	SH'T NO.	REV.
		01 / 01	0

[illegible]

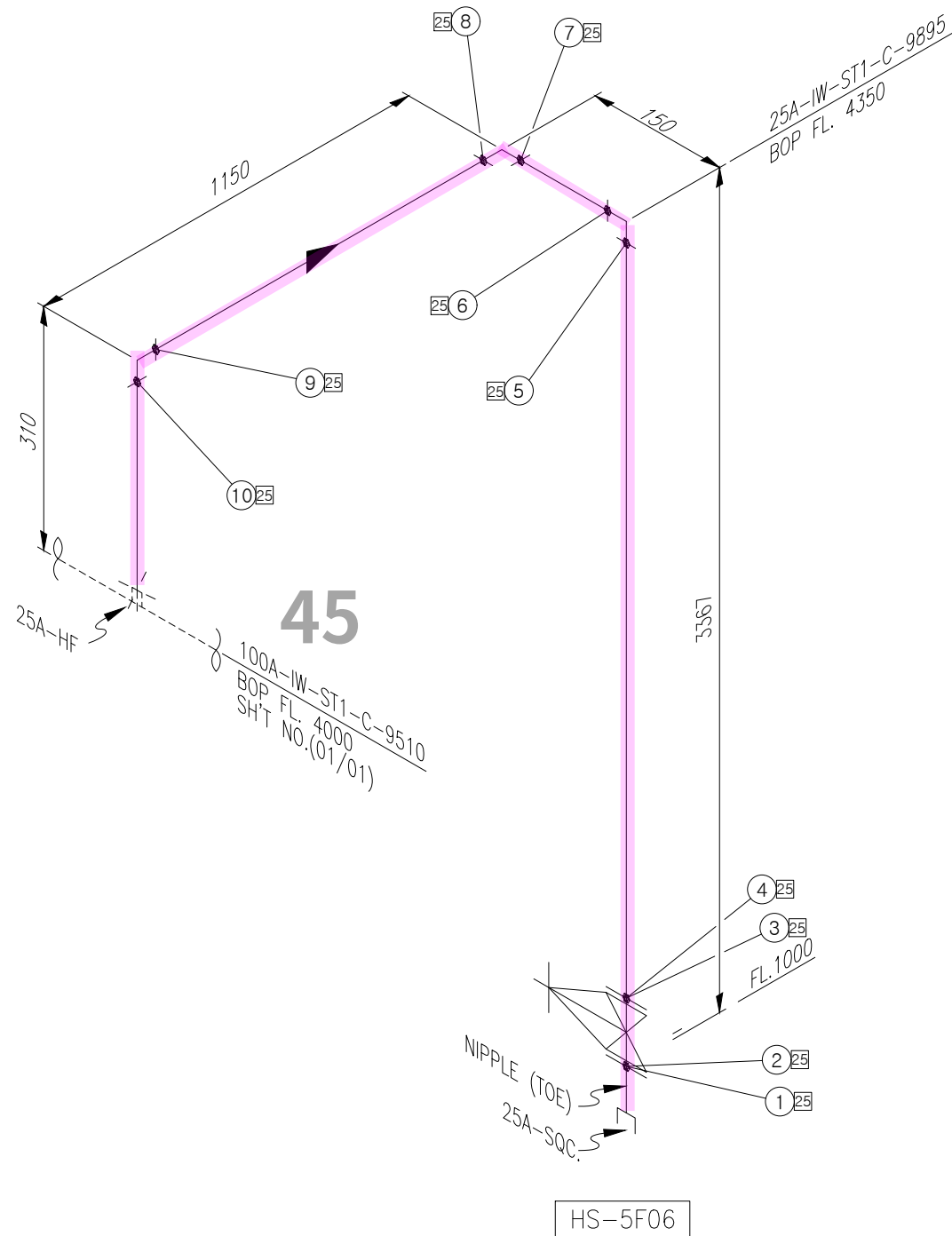
<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA									M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	COMMODITY	IW	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000081-A							REV. NO.	DATE	DESCRIPTION	BY



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	4.7M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	3EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLC'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)
IN	25tX25A	4.7M	관 보온재(고무발포)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT	5FA								M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	IW	JOB NO.
						REF. P & ID CJ-M15W13-PR-A20-000081-A								LINE NO.	25A-IW-ST1-C-9895	SH'T NO. 01 / 01
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 외부에 유출·제3자에게 공개, 누설 또는 사용·포지하는 행위를 금지하며, 회사 내부 직원도 이를 위반할 경우 법적 책임을 지게 됩니다.



CLIENT 20

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY IW		JOB NO. 22330D	
LINE NO. 40A-IW-ST1-C-9525		SH'T NO. 01 / 06	REV. 0

[illegible]

<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

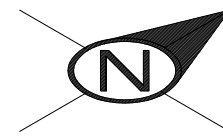
CLIENT



21

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR 	PROJECT NAME					
	THICK.	25	DESIGN	13.5	60	UNIT 5FA							M15X Ph4 Project					
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	IW	JOB NO.			
																		22330D
						REF. P & ID CJ-M15W14-PR-A20-000142							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

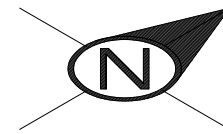


1. (*) FIELD ADJUST

[illegible]

CLIENT		22
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INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	13.5	60	UNIT 5FA								M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A								COMMODITY	IW	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000142		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	40A-IW-ST1-C-9525	SH'T NO. 03 / 06	REV. 0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



1. (*) FIELD ADJUST

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	40A-IW-ST1-C-9525	SH'T NO.	REV.
		05 / 06	0

[illegible]

<NOTE>

1. (*) FIELD ADJUST

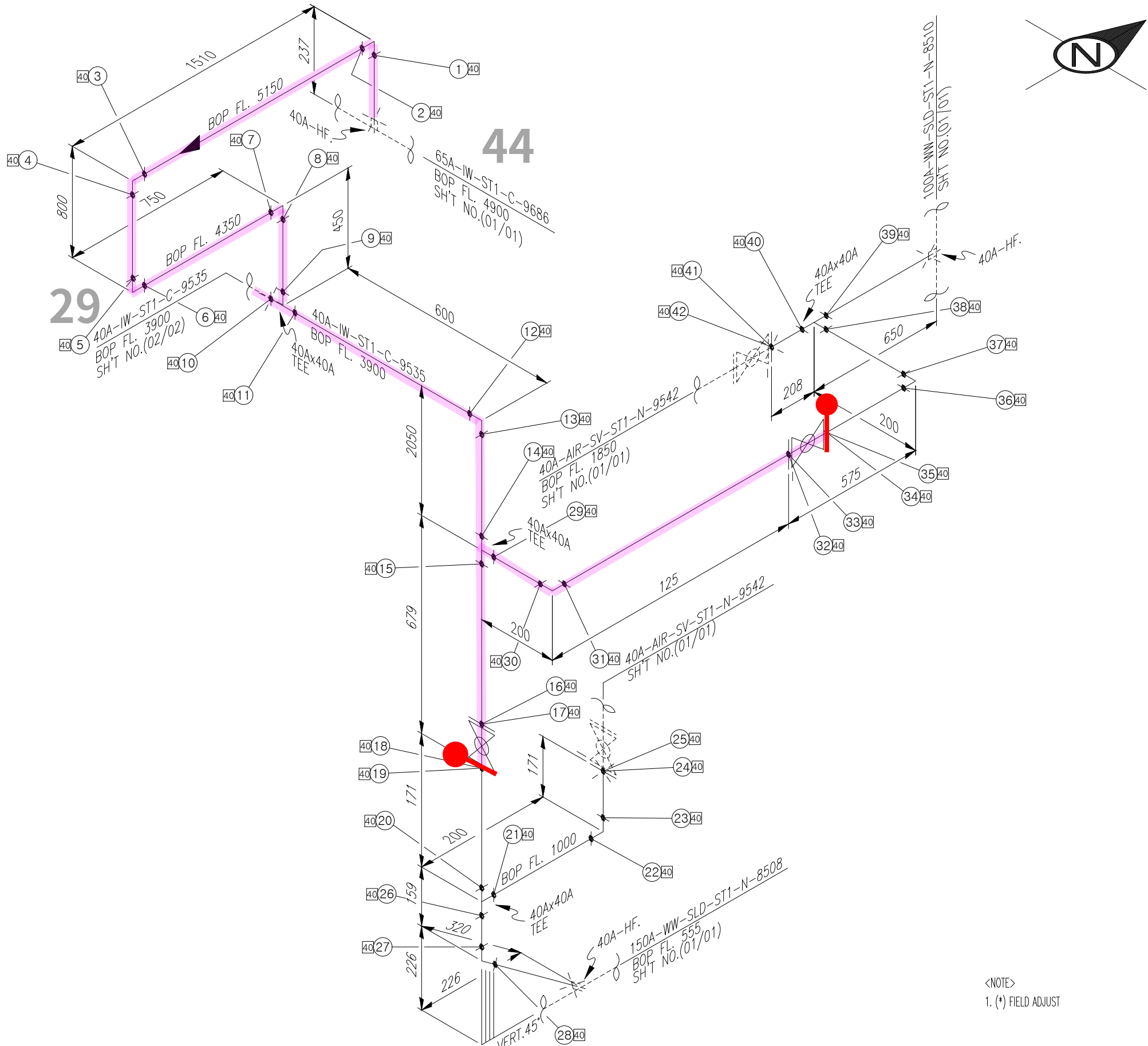
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA									M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	IW	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000142							REV. NO.	DATE	DESCRIPTION	BY

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



1. (*) FIELD ADJUST



<NOTE>
1. (*) FIELD ADJUST

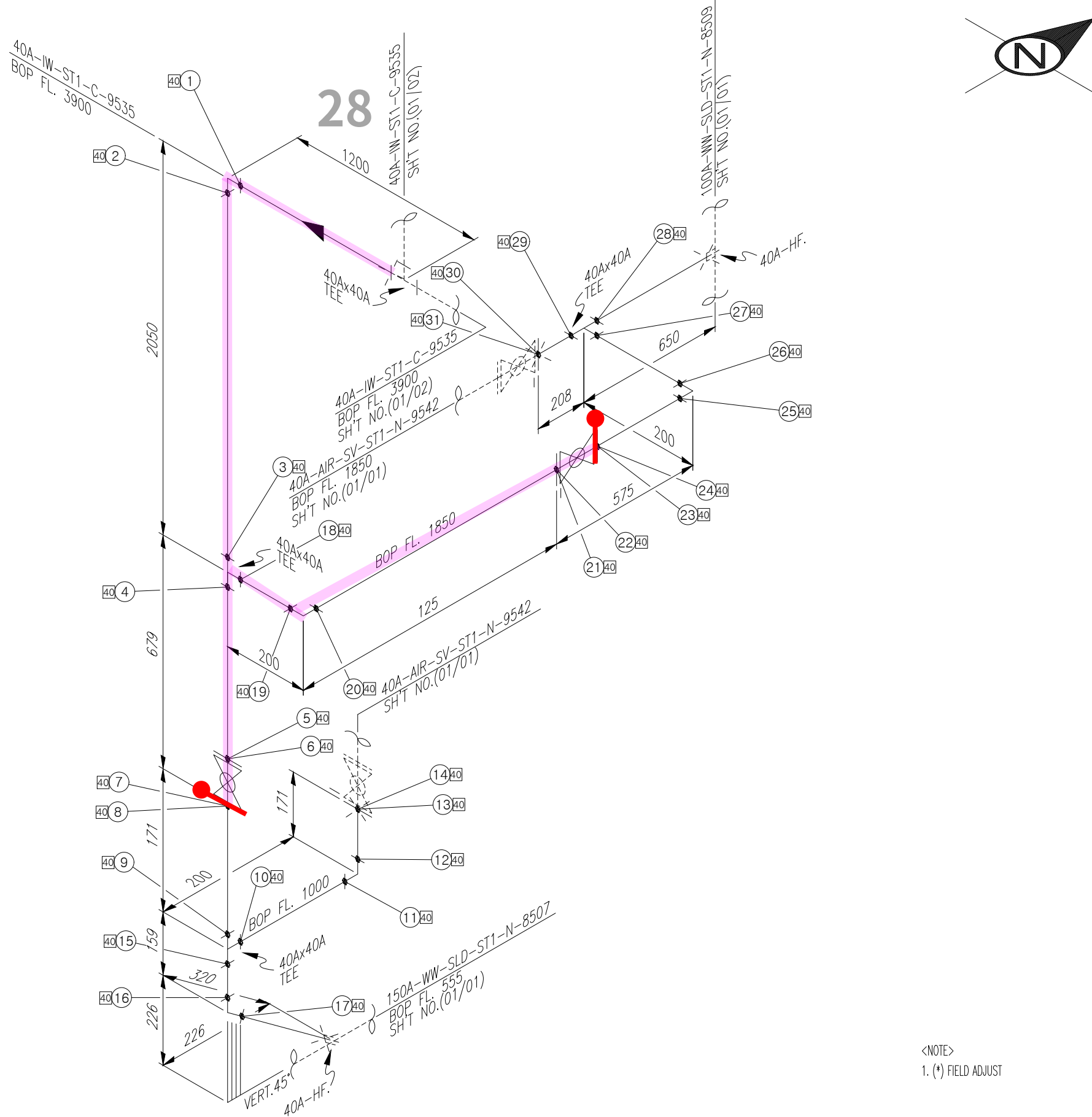
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR 	PROJECT NAME			
	THICK.	25	DESIGN	13.5	60	UNIT	5FA										M15X Ph4 Project			
				OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D										COMMODITY	IW	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000103		0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영				LINE NO.	40A-IW-ST1-C-9535	SH'T NO. 01 / 02	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D							

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



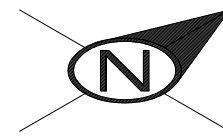
28



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT	5FA										
			OPERATING	10	AMB	REF. PLAN DWG.		△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-050007-D											
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			LINE NO.	SH'T NO.
						CJ-M15W14-PR-A20-000103											
																40A-IW-ST1-C-9535	02 / 02
																	REV. 0

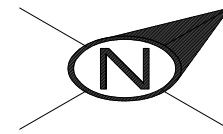


* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* SUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL-PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050006-C						
						REF. P & ID CJ-M15W14-PR-A20-000104		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

CLIENT  30

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY IW		JOB NO. 22330D	
LINE NO. 40A-IW-ST1-C-9536		SH'T NO. 01 / 02	REV. 0



1. (*) FIELD ADJUST

[illegible]

CLIENT



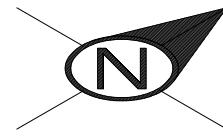
31

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	40A-IW-ST1-C-9536	SH'T NO.	REV.
		02 / 02	0

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/G1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA								M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050006-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	JOB NO.	
						IW							22330D		
						REF. P & ID CJ-M15W14-PR-A20-000104	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
													40A-IW-ST1-C-9536	02 / 02	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

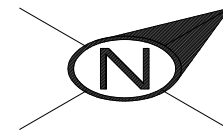


1. (*) FIELD ADJUST

CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	40A-IW-ST1-C-9537	SH'T NO. 02 / 02	REV. 0

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000105						
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



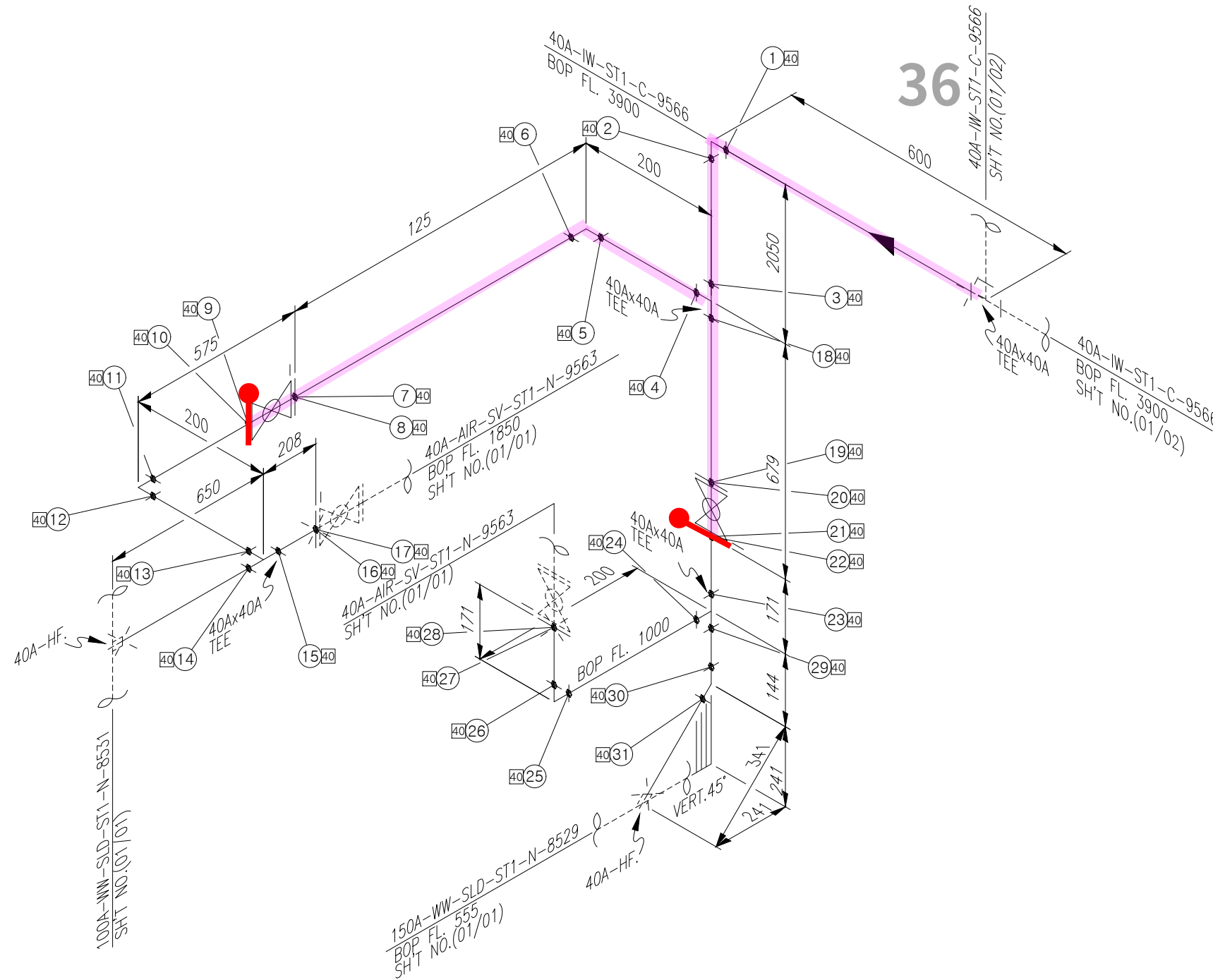
<NOTE>
1. (*) FIELD ADJUST

CLIENT  34

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	40A	7.0M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
E45	40A	1EA	ELBOW 45 DEG, STSF304, SW – SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW – SCH 10S
VB	40A	2EA	BALL VALVE, KS 10K, RF FLC'D, BODY–STSF304 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
F	40A	6EA	FLC SO, KS 10K, RF, STS304
G	40A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
IN	25tX40A	7.0M	관 보온재(고무발포)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/G1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA								M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D								COMMODITY	JOB NO.
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		22330D	
						REF. P & ID CJ-M15W14-PR-A20-000108	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	40A-IW-ST1-C-9566	SH'T NO. 02 / 02

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

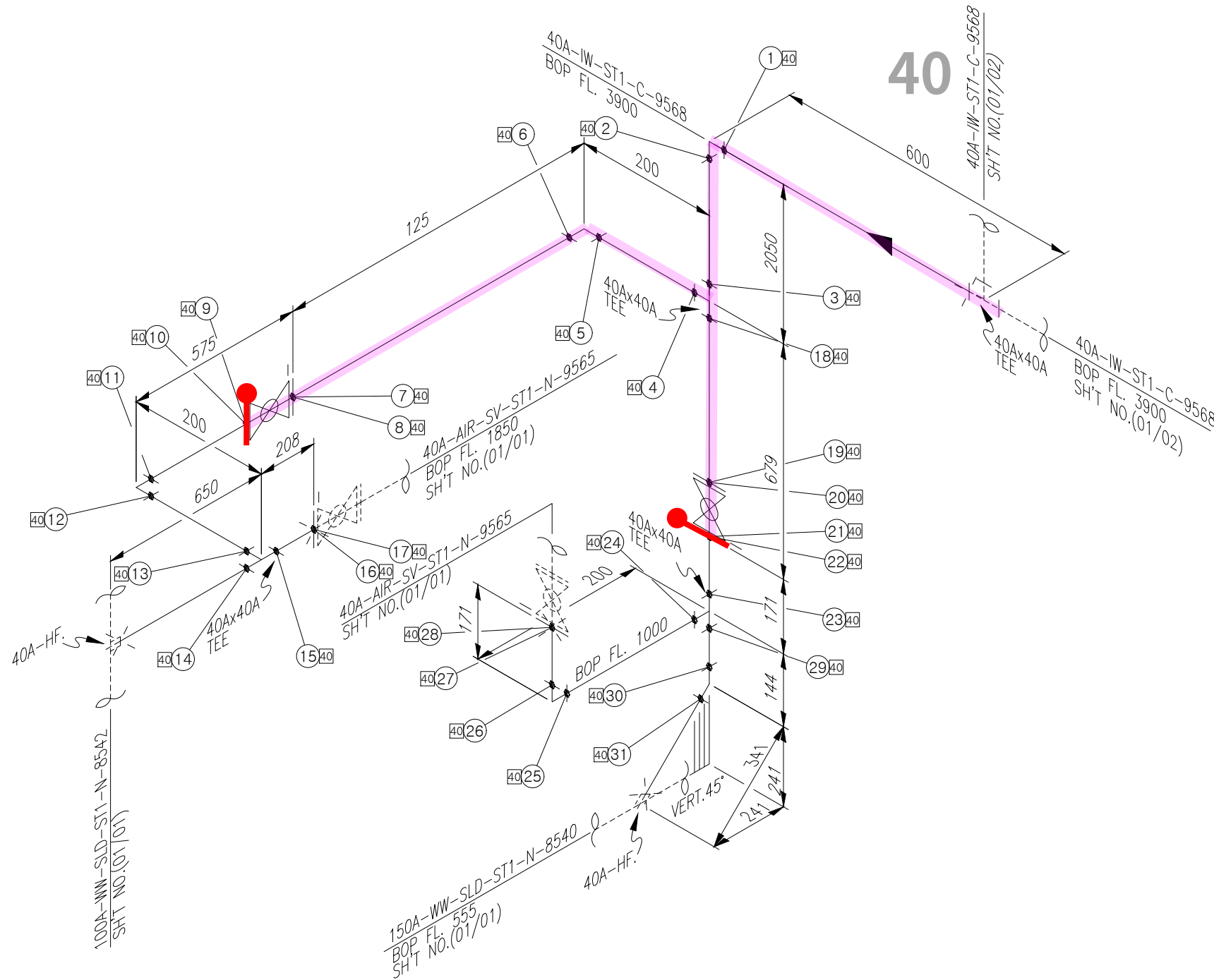


CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY IW		JOB NO. 22330D	
LINE NO. 40A-IW-ST1-C-9567		SH'T NO. 01 / 02	REV. 0

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C						
						REF. P & ID CJ-M15W14-PR-A20-000109		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	40A	7.0M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
E45	40A	1EA	ELBOW 45 DEG, STSF304, SW – SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW – SCH 10S
VB	40A	2EA	BALL VALVE, KS 10K, RF FLC'D, BODY–STSF304 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
F	40A	6EA	FLG SO, KS 10K, RF, STS304
G	40A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
IN	25tX40A	7.0M	관 보온재(고무발포)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT	5FA								M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-050005-D								IW	22330D	
						REF. P & ID								LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000110								40A-IW-ST1-C-9568	02 / 02	0
									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

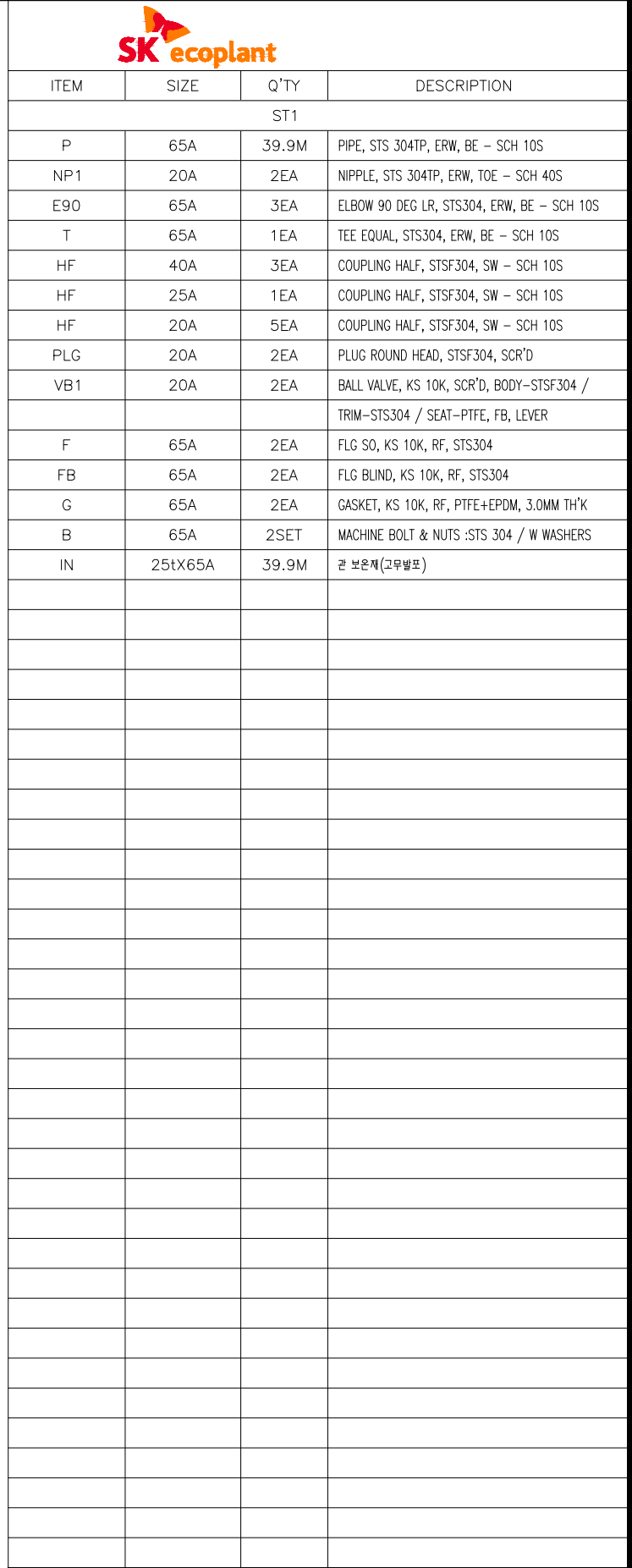
경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

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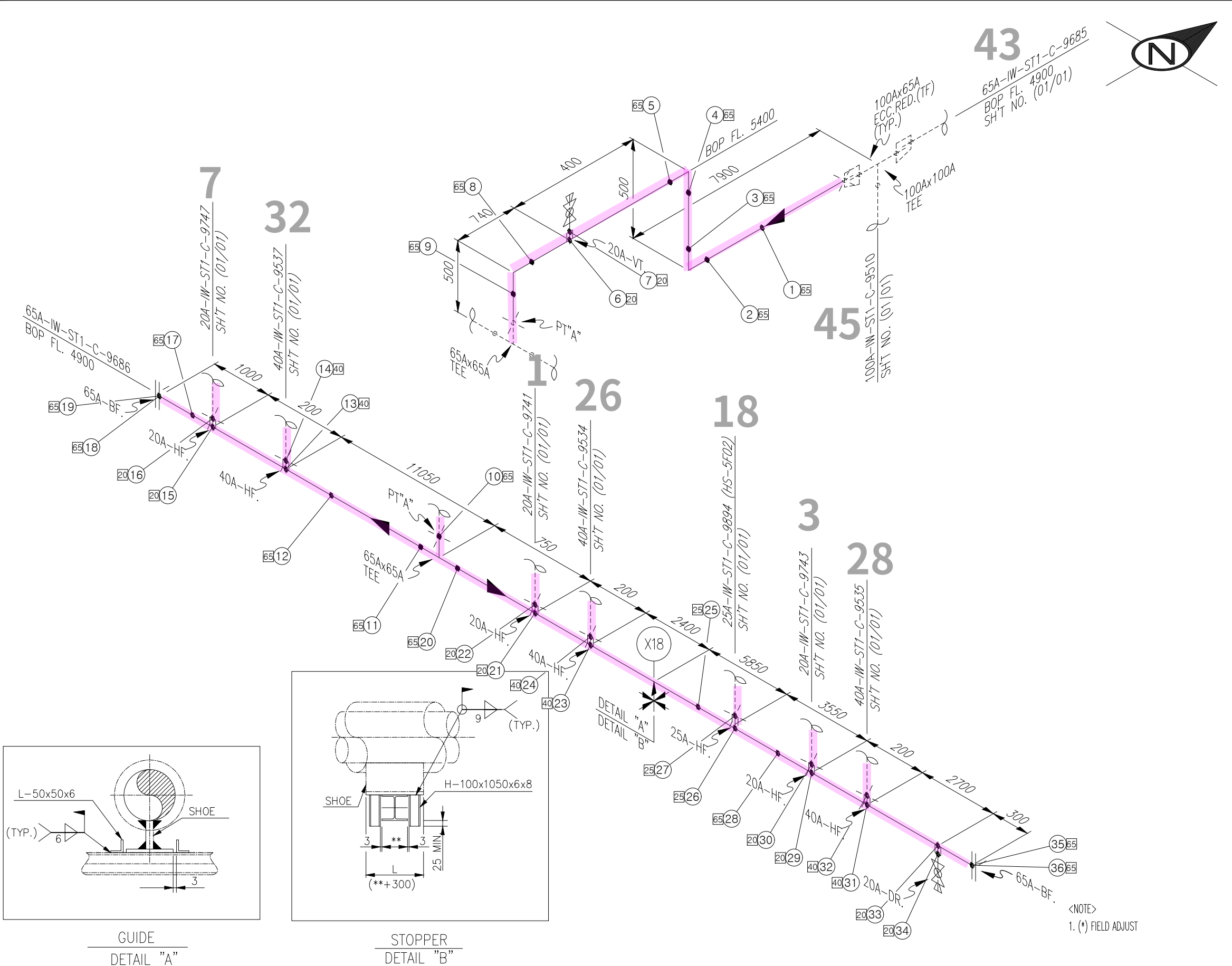
42

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA							M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C						COMMODITY	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000128		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO. 01 / 01
												50A-IW-ST1-C-9581		

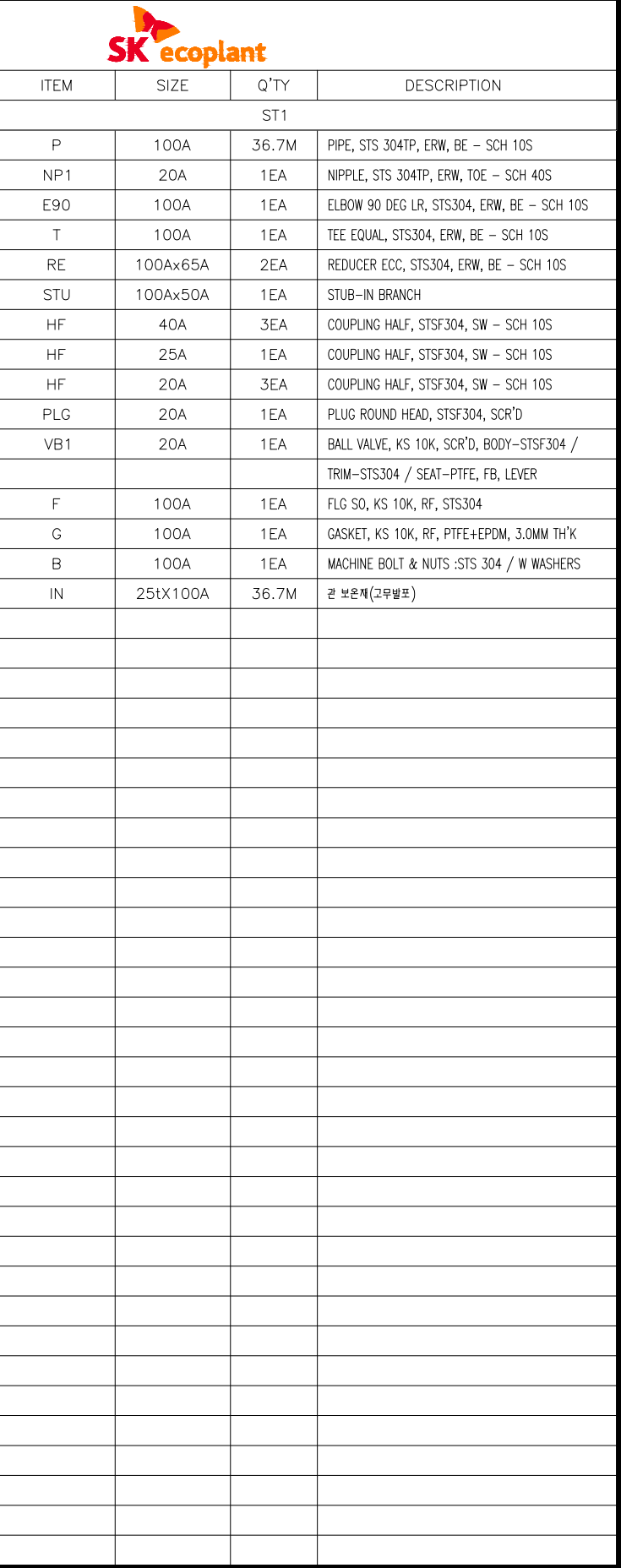


CLIENT		 43	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	65A-IW-ST1-C-9685	SH'T NO.	REV.
		01 / 01	0

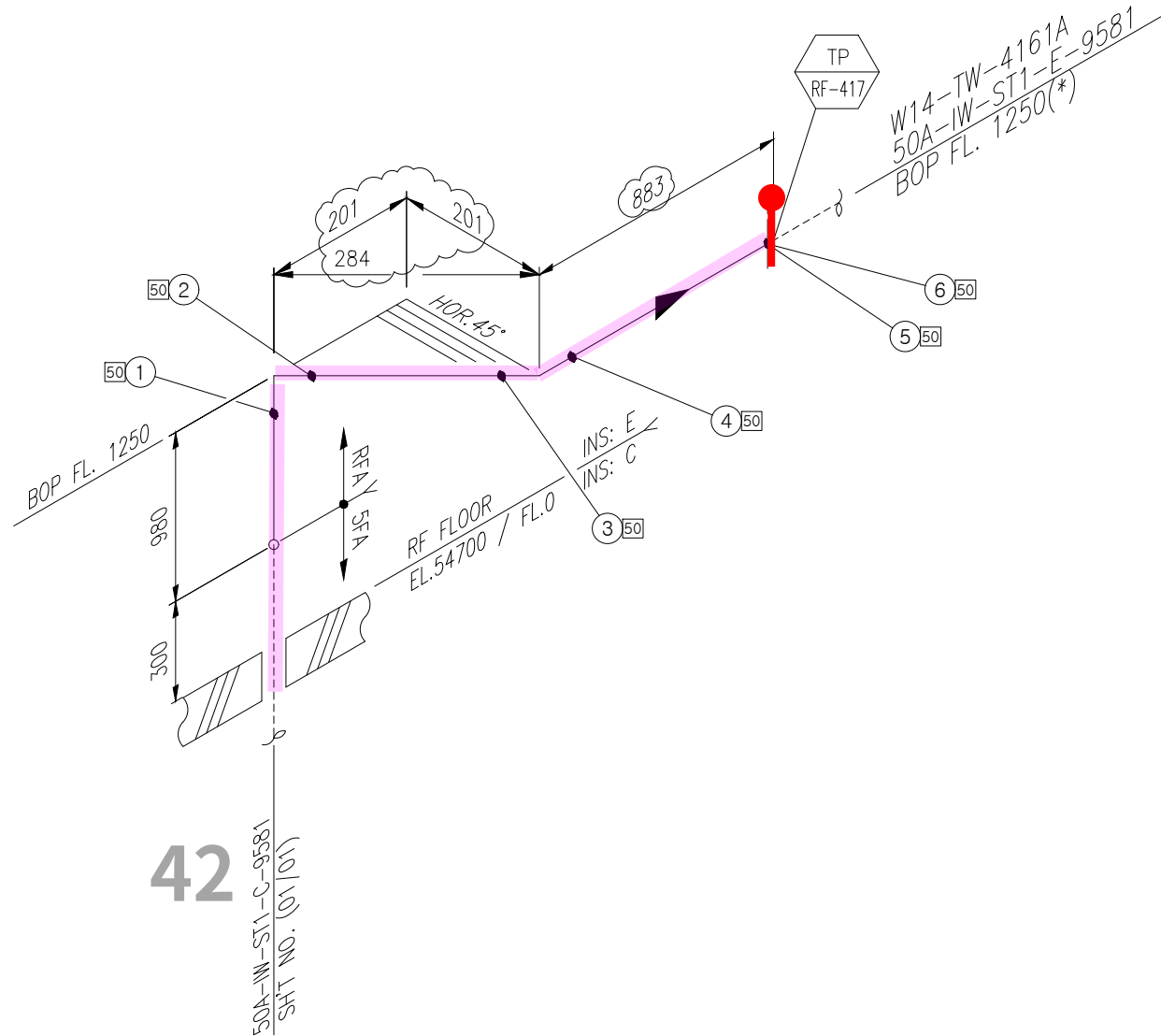


* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR		PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT	5FA									M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG.								COMMODITY	IW	JOB NO.	
						CJ-M15W14-PI-A30-050009-D		0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	65A-IW-ST1-C-9686	SH'T NO.	REV.
						CJ-M15W13-PR-A20-000081-A										01 / 01	0



CLIENT				45	
CONTRACTOR			PROJECT NAME		
			M15X Ph4 Project		
COMMODITY			JOB NO.		
IW			22330D		
LINE NO.			SH'T NO.		REV.
100A-IW-ST1-C-9510			01 / 01		0

[illegible]

<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
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[illegible]

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음